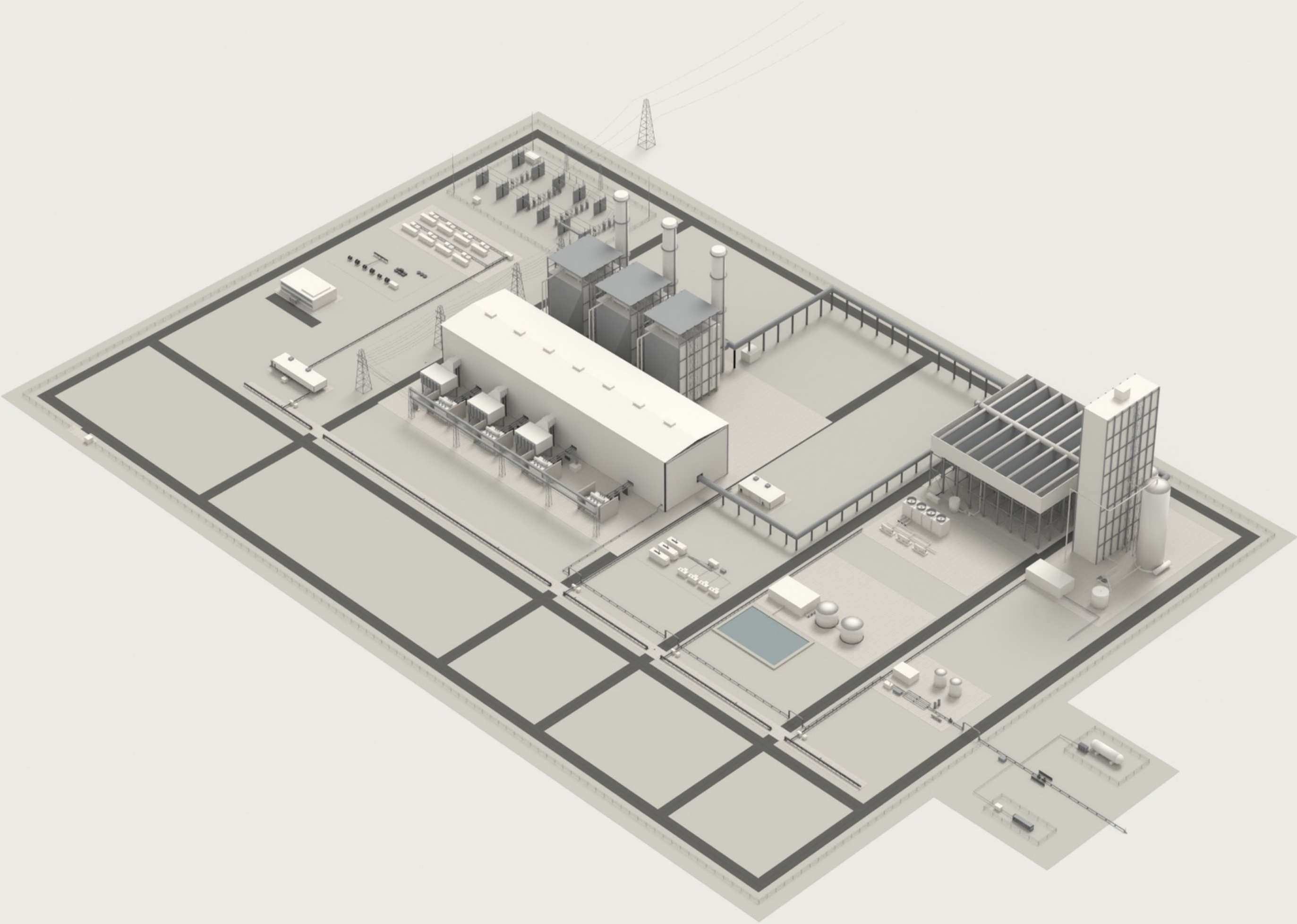


Whole site

Site arrangement | the sixteen zones and how the power block sits between fuel, cooling and the grid

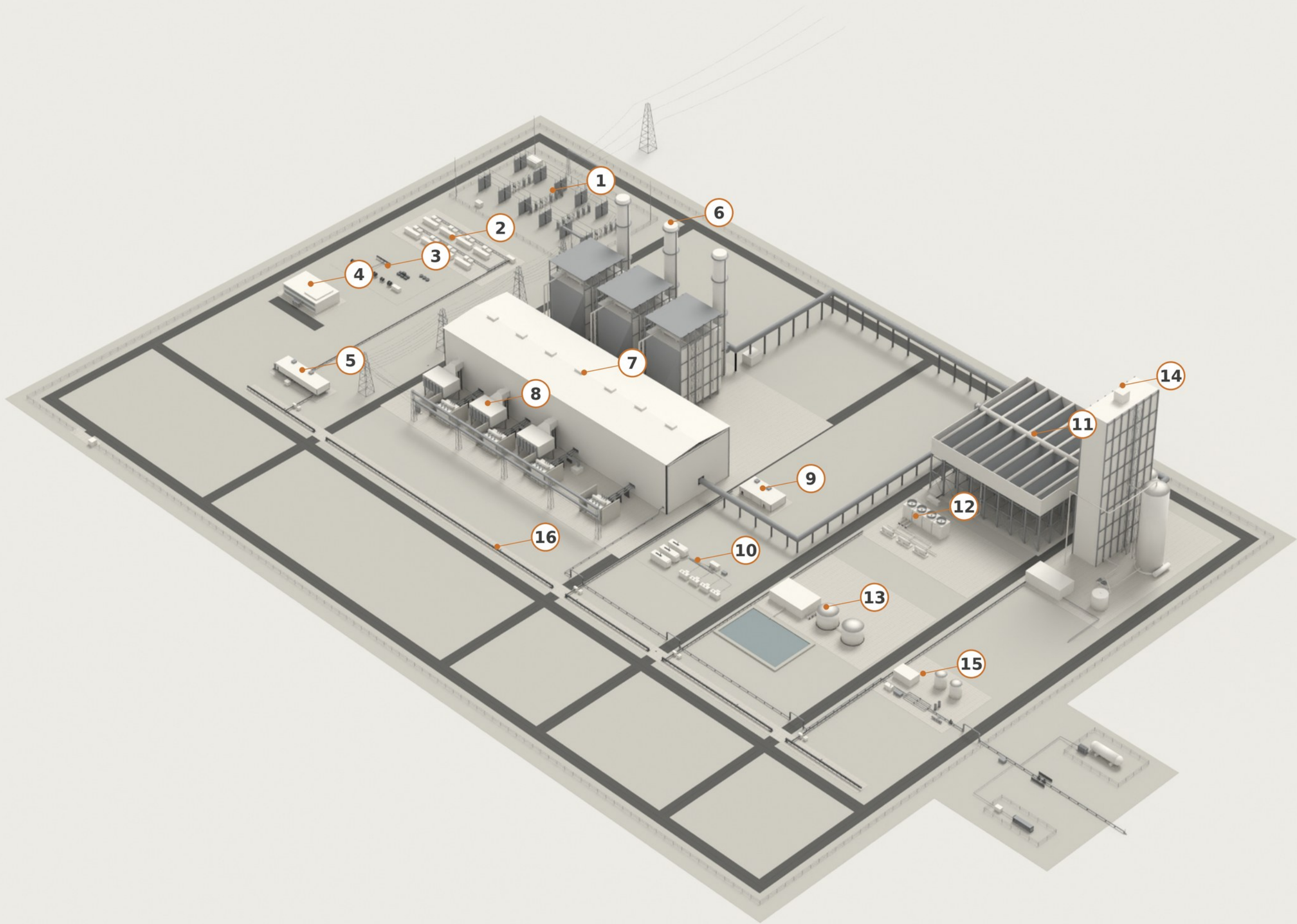


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Explains where each package sits and which way the gas, steam, cooling and electrical paths run across the site.
Illustrative arrangement. Generic equipment and circuits; not procurement or construction data.

Whole site

Site arrangement | the sixteen zones and how the power block sits between fuel, cooling and the grid



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- ① Switchyard and GSU transformers

② BESS yard

③ Reel staging, laydown and prefab spine

④ Admin building and control room
- ⑤ Elevated modular e-house

⑥ HRSGs and stacks

⑦ Turbine hall (3 GT + 1 ST)

⑧ GT air-inlet filter houses
- ⑨ MCC / VFD / UPS building

⑩ Modular power yard

⑪ Air-cooled condenser

⑫ Chillers and auxiliary cooling towers
- ⑬ Water treatment, tanks and pond

⑭ Carbon capture block

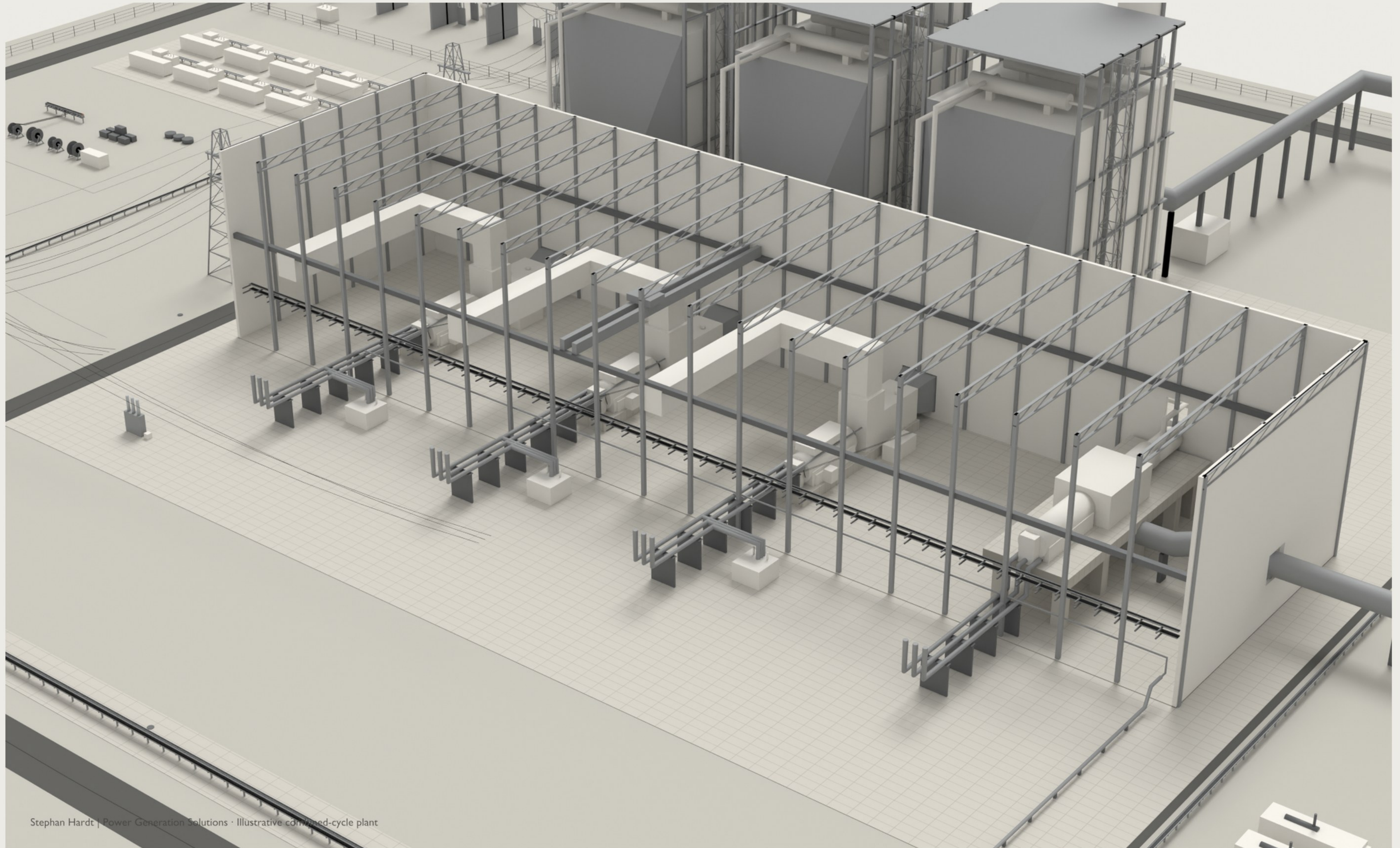
⑮ Gas metering and process tanks

⑯ Site cable corridor and MV distribution

Explains where each package sits and which way the gas, steam, cooling and electrical paths run across the site. Illustrative arrangement. Generic equipment and circuits; not procurement or construction data.

Turbine hall cutaway

GT train | gas turbine, generator, generator leads and isophase bus to the GSU

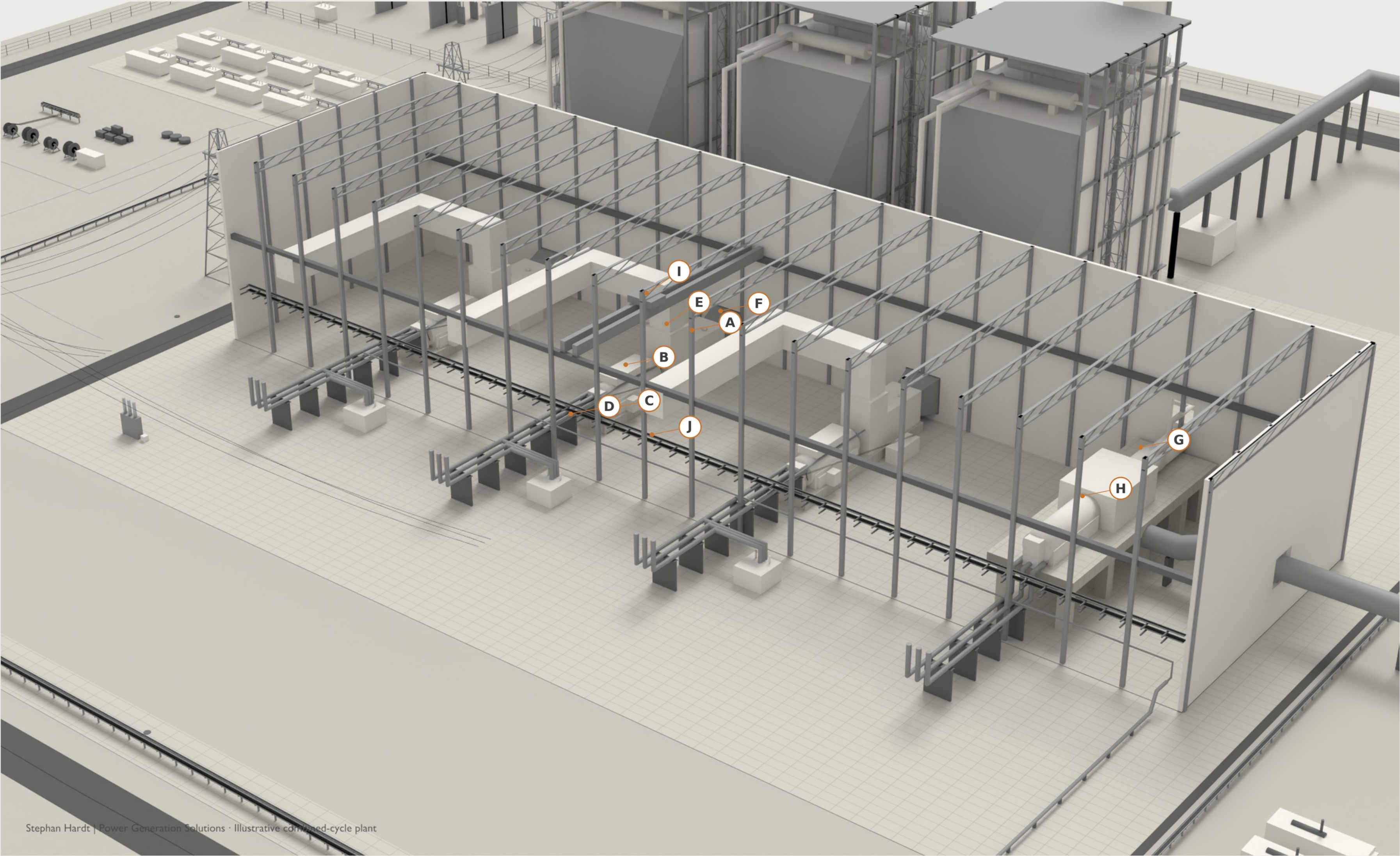


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Explains the generator connection: leads at the terminal box, isophase bus out through the wall, and the tray tiers that carry auxiliaries. Illustrative arrangement. Generic equipment and circuits; not procurement or construction data.

Turbine hall cutaway

GT train | gas turbine, generator, generator leads and isophase bus to the GSU

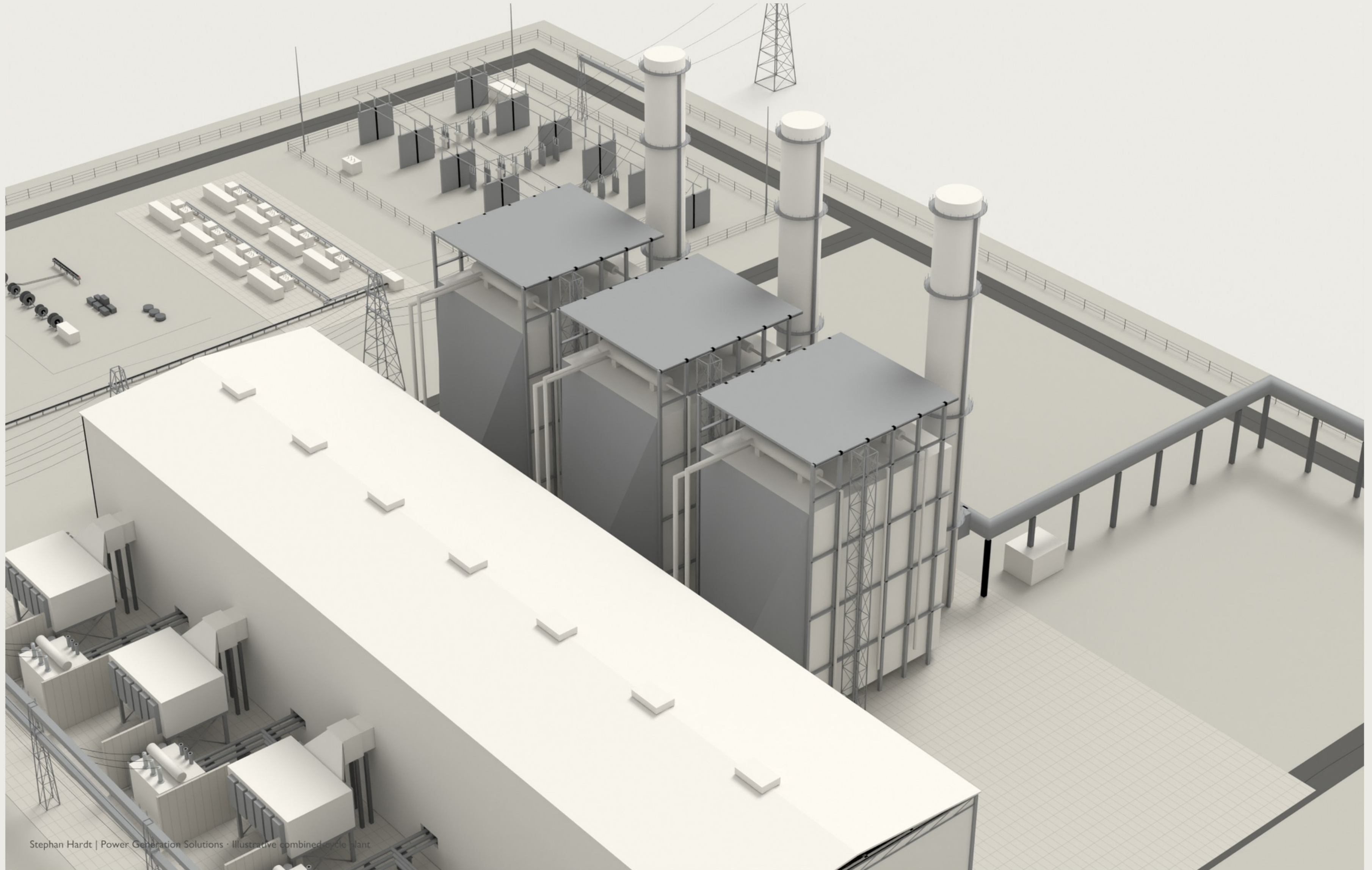


- A** Gas turbine package (enclosed)
- B** GT generator
- C** Generator leads and terminal box
- D** Isophase bus duct (generator to GSU)
- E** Inlet plenum and silencer
- F** Exhaust diffuser to HRSG
- G** Steam turbine (HP / IP / LP)
- H** Steam turbine generator
- I** Turbine hall bridge crane
- J** Hall cable tray (power below, controls above)

Explains the generator connection: leads at the terminal box, isophase bus out through the wall, and the tray tiers that carry auxiliaries. Illustrative arrangement. Generic equipment and circuits; not procurement or construction data.

HRSG and stack

Heat recovery | exhaust from the GT through the HRSG casing to the stack, with the steam headers to the ST

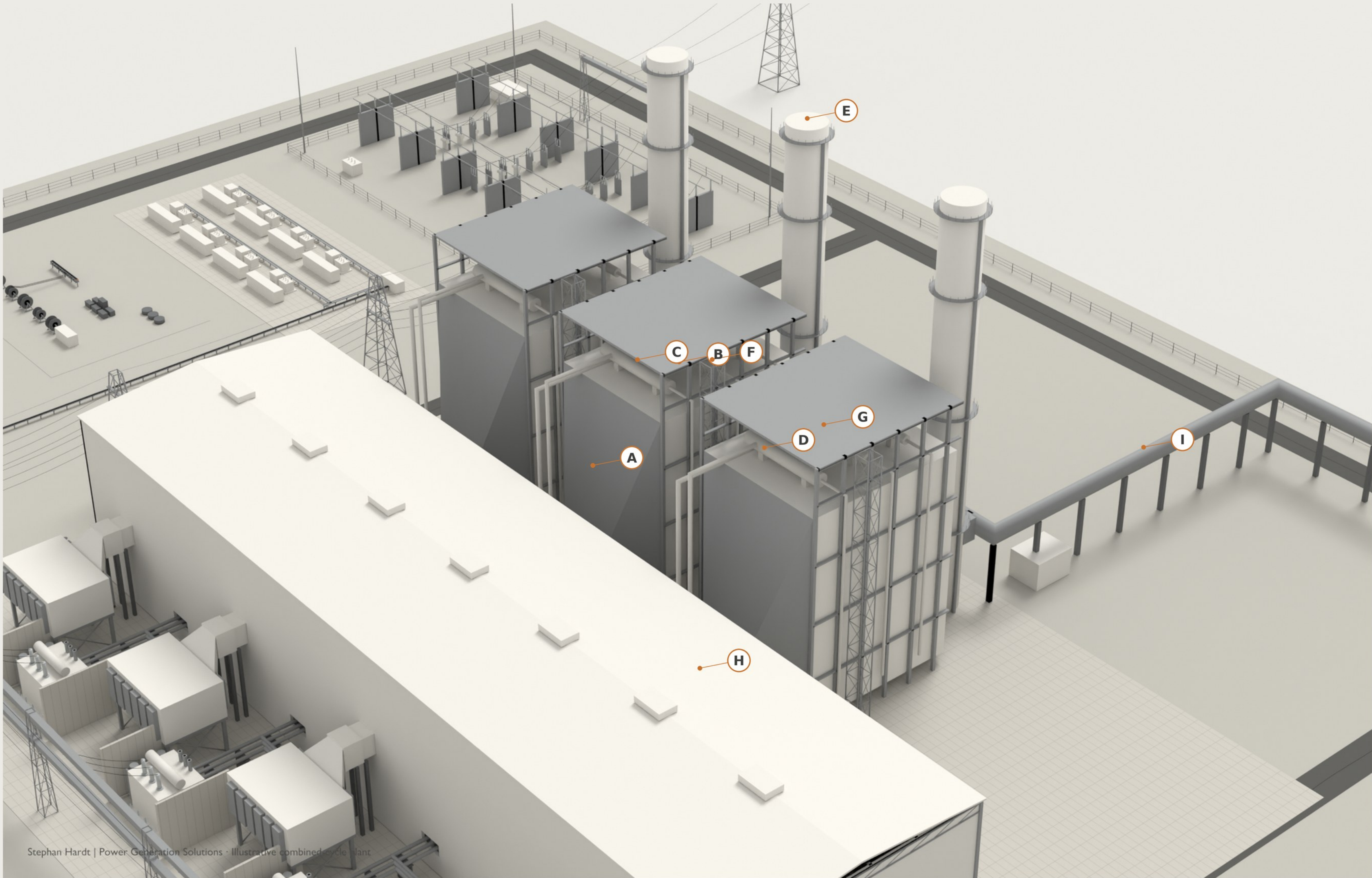


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Explains the gas path from turbine exhaust to stack and where the flue gas can be diverted to the capture block.
Illustrative arrangement. Generic equipment and circuits; not procurement or construction data.

HRSG and stack

Heat recovery | exhaust from the GT through the HRSG casing to the stack, with the steam headers to the ST

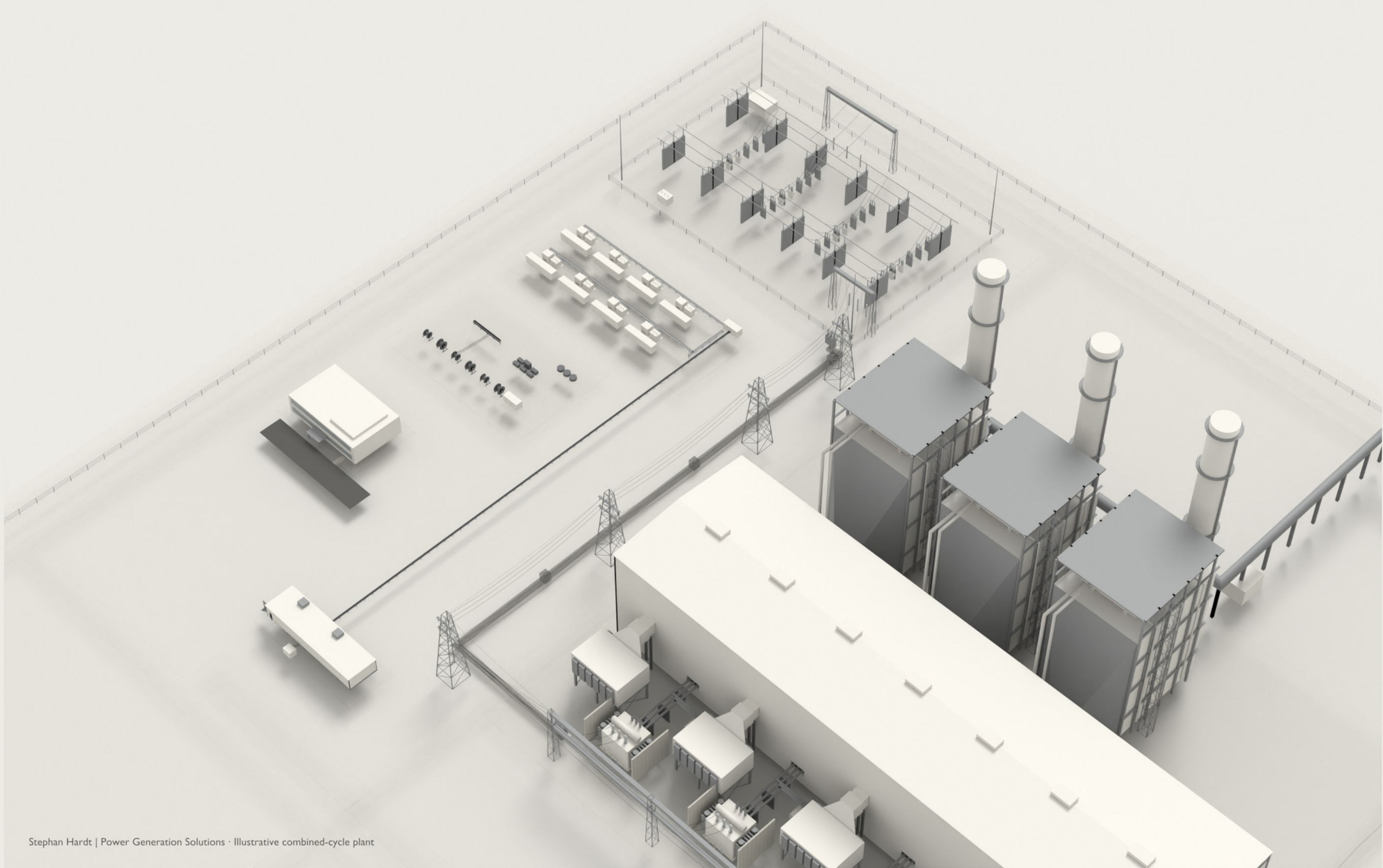


- A** Inlet transition duct from GT exhaust
- B** HRSG casing (triple-pressure, reheat)
- C** Steam drums
- D** Outlet transition to stack
- E** HRSG exhaust stack
- F** Access stair tower
- G** Flue-gas take-off and damper to CCS
- H** HP and reheat steam headers (3 HRSGs to 1 ST)
- I** Flue-gas collector duct from stacks

Explains the gas path from turbine exhaust to stack and where the flue gas can be diverted to the capture block.
Illustrative arrangement. Generic equipment and circuits; not procurement or construction data.

Switchyard and GSUs

Power export | GSU transformers, the overhead gen-tie and the underground HV getaway option (ground shown X-ray)

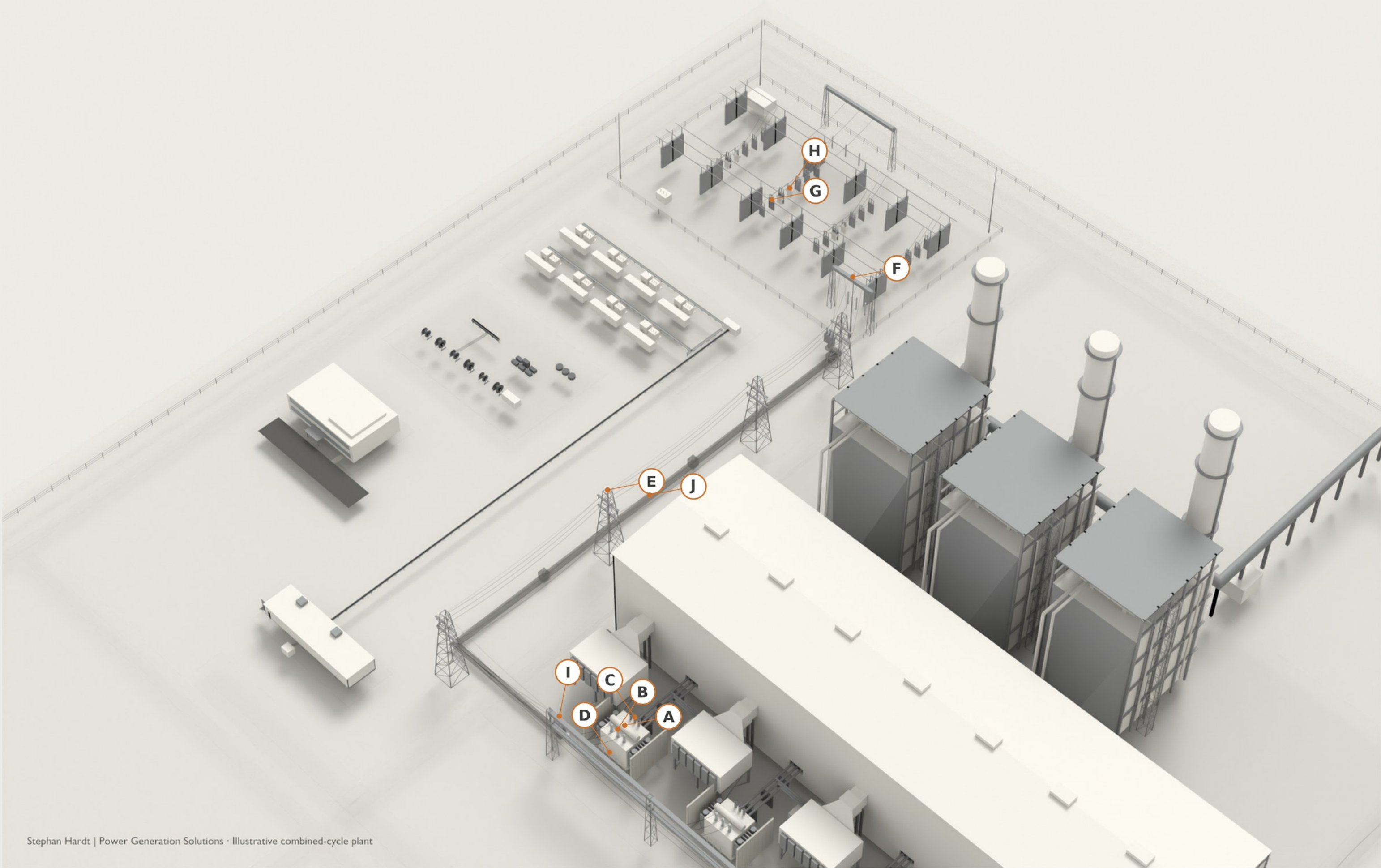


Stephan Hardt | Power Generation Solutions · Illustrative combined-cycle plant

Explains the two ways the 230 kV connection can leave the GSUs: overhead on towers, or underground in a duct bank with sealing ends at each end. Illustrative arrangement. Generic equipment and circuits; not procurement or construction data.

Switchyard and GSUs

Power export | GSU transformers, the overhead gen-tie and the underground HV getaway option (ground shown X-ray)

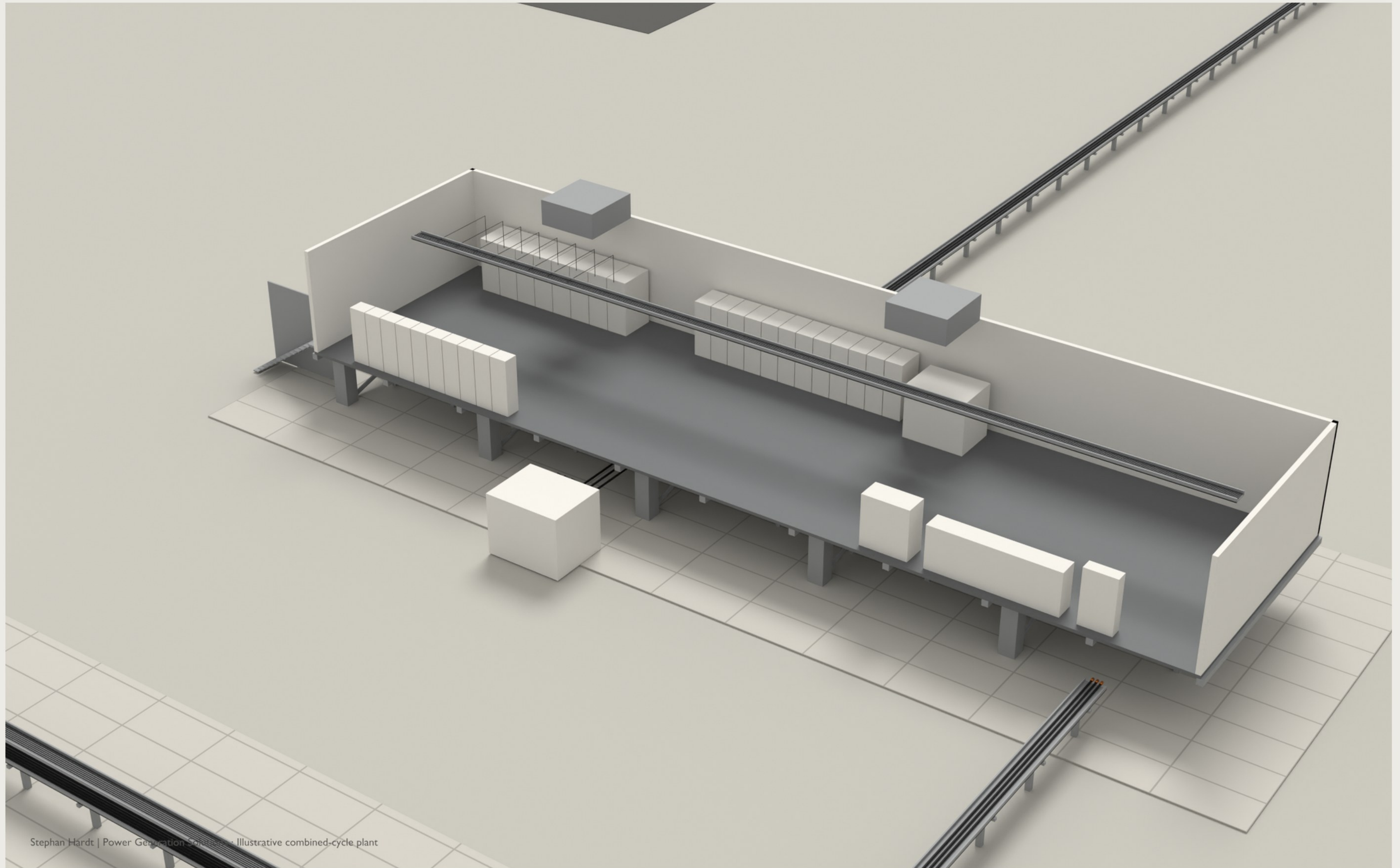


- A** GT-1 generator step-up transformer (18/230 kV)
- B** 230 kV HV bushings
- C** LV bushings (IPB termination)
- D** Surge arresters
- E** 230 kV double-circuit gen-tie towers
- F** Gen-tie dead-end structure
- G** 230 kV main buses
- H** 230 kV circuit breakers
- I** Underground option: cable sealing ends (plant end)
- J** Underground option: 230 kV XLPE duct bank (below grade)

Explains the two ways the 230 kV connection can leave the GSUs: overhead on towers, or underground in a duct bank with sealing ends at each end. Illustrative arrangement. Generic equipment and circuits; not procurement or construction data.

E-house cutaway

MV collection and auxiliaries | elevated modular e-house with the cable vault beneath

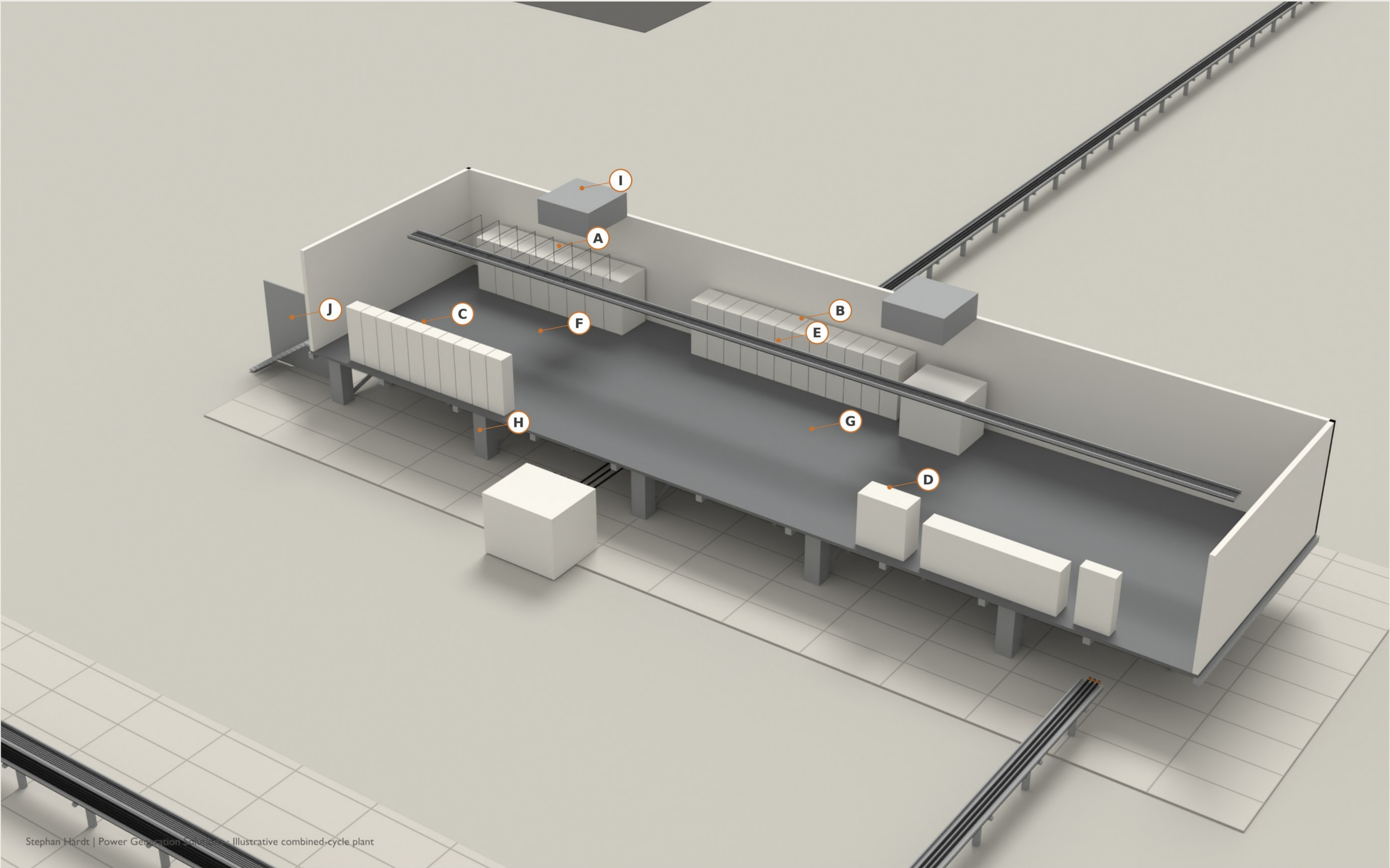


Stephan Hardt | Power Generation Solutions | Illustrative combined-cycle plant

Explains how feeders arrive below the floor, rise into the switchgear and how controls run overhead to the relay panels.
Illustrative arrangement. Generic equipment and circuits; not procurement or construction data.

E-house cutaway

MV collection and auxiliaries | elevated modular e-house with the cable vault beneath

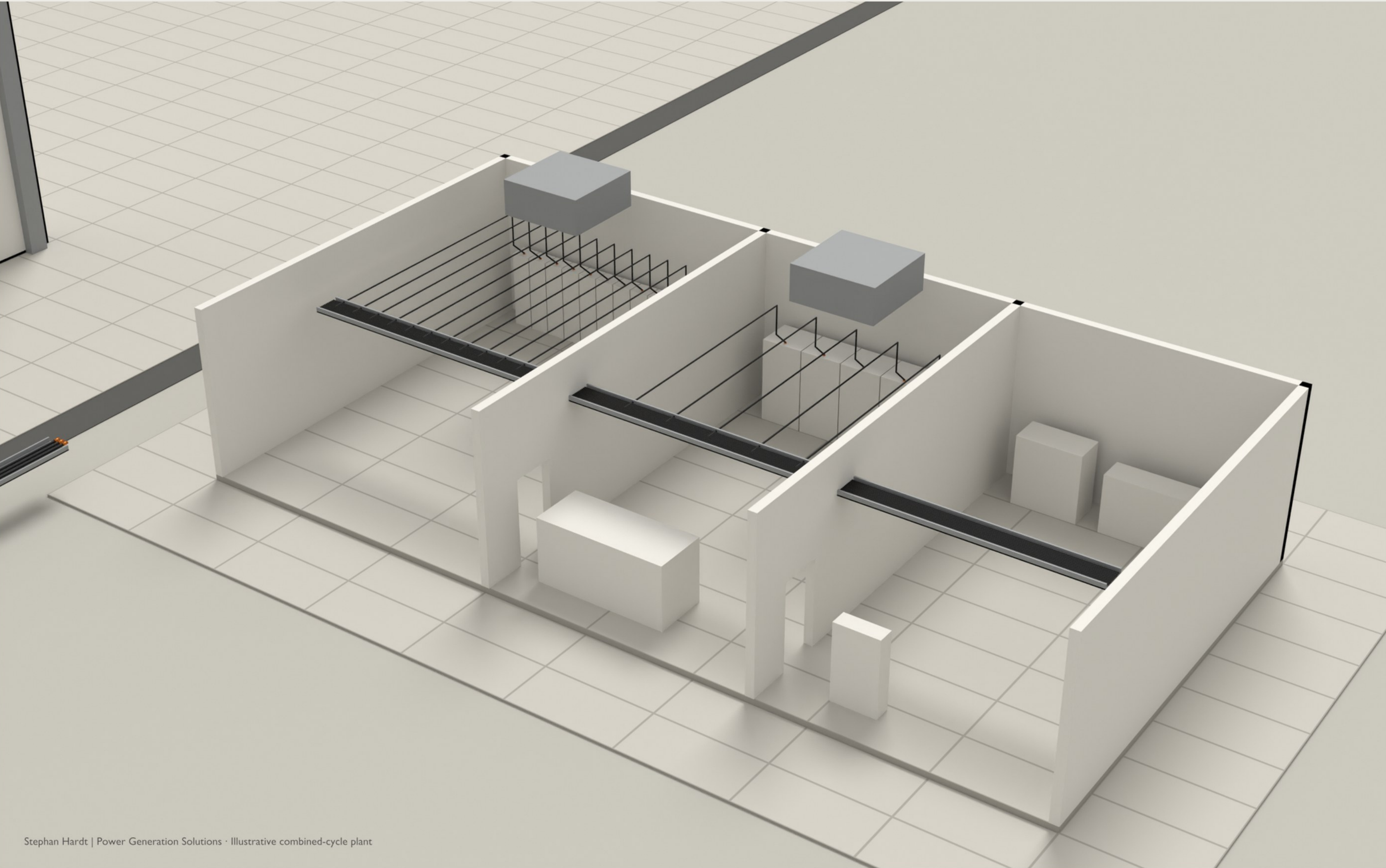


- A MV switchgear lineup (13.8 kV)
- B LV switchgear (480 V)
- C Protection and relay panels
- D UPS and battery rack
- E Overhead control tray
- F Power cable risers from cable vault
- G Under-floor cable tray (vault)
- H Elevated support steel
- I Roof HVAC units
- J Access stair and landing

Explains how feeders arrive below the floor, rise into the switchgear and how controls run overhead to the relay panels. Illustrative arrangement. Generic equipment and circuits; not procurement or construction data.

MCC / VFD / UPS room cutaway

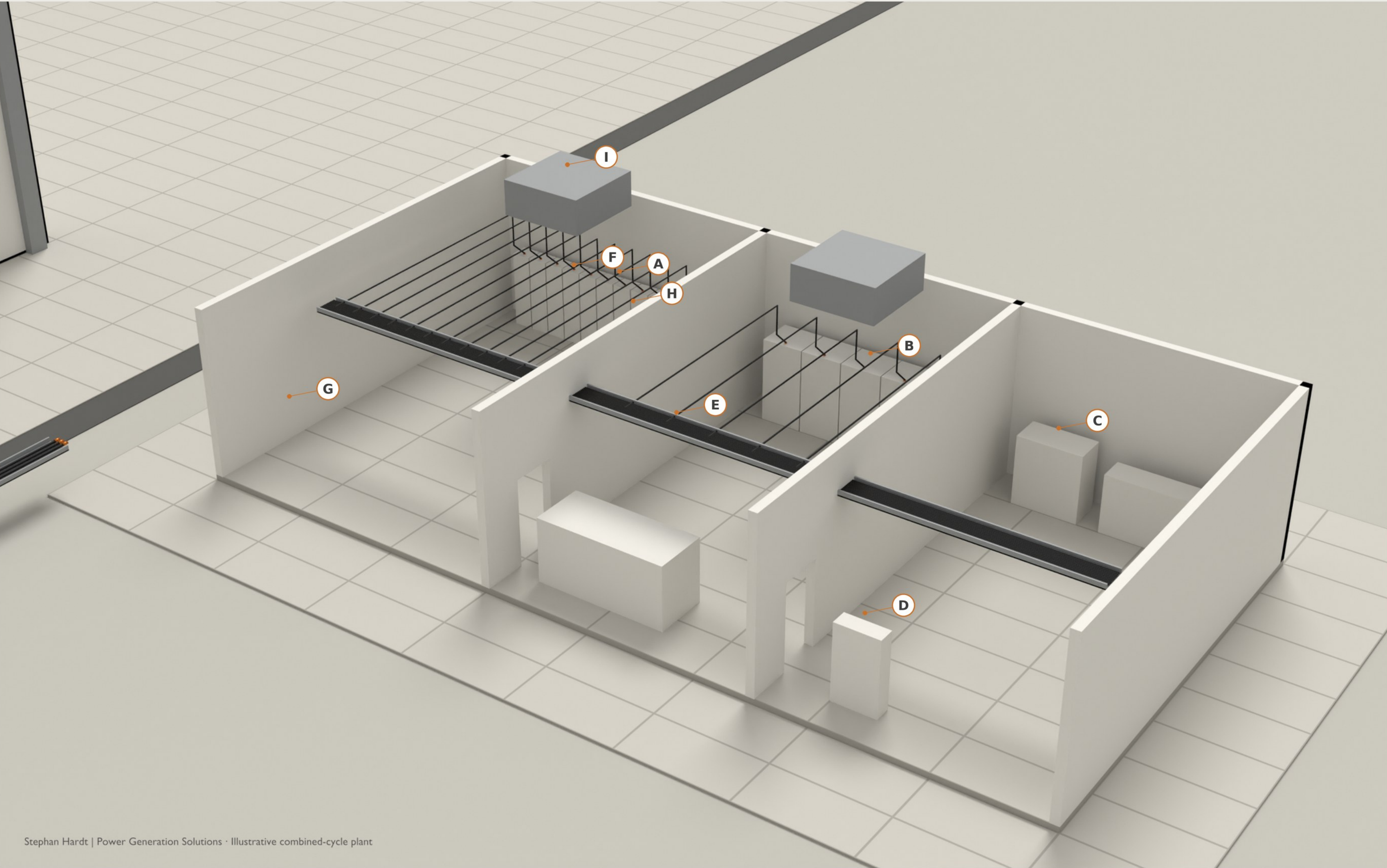
Motor control | MCC, drives and UPS in separated rooms with a common overhead tray



Stephan Hardt | Power Generation Solutions · Illustrative combined-cycle plant

MCC / VFD / UPS room cutaway

Motor control | MCC, drives and UPS in separated rooms with a common overhead tray

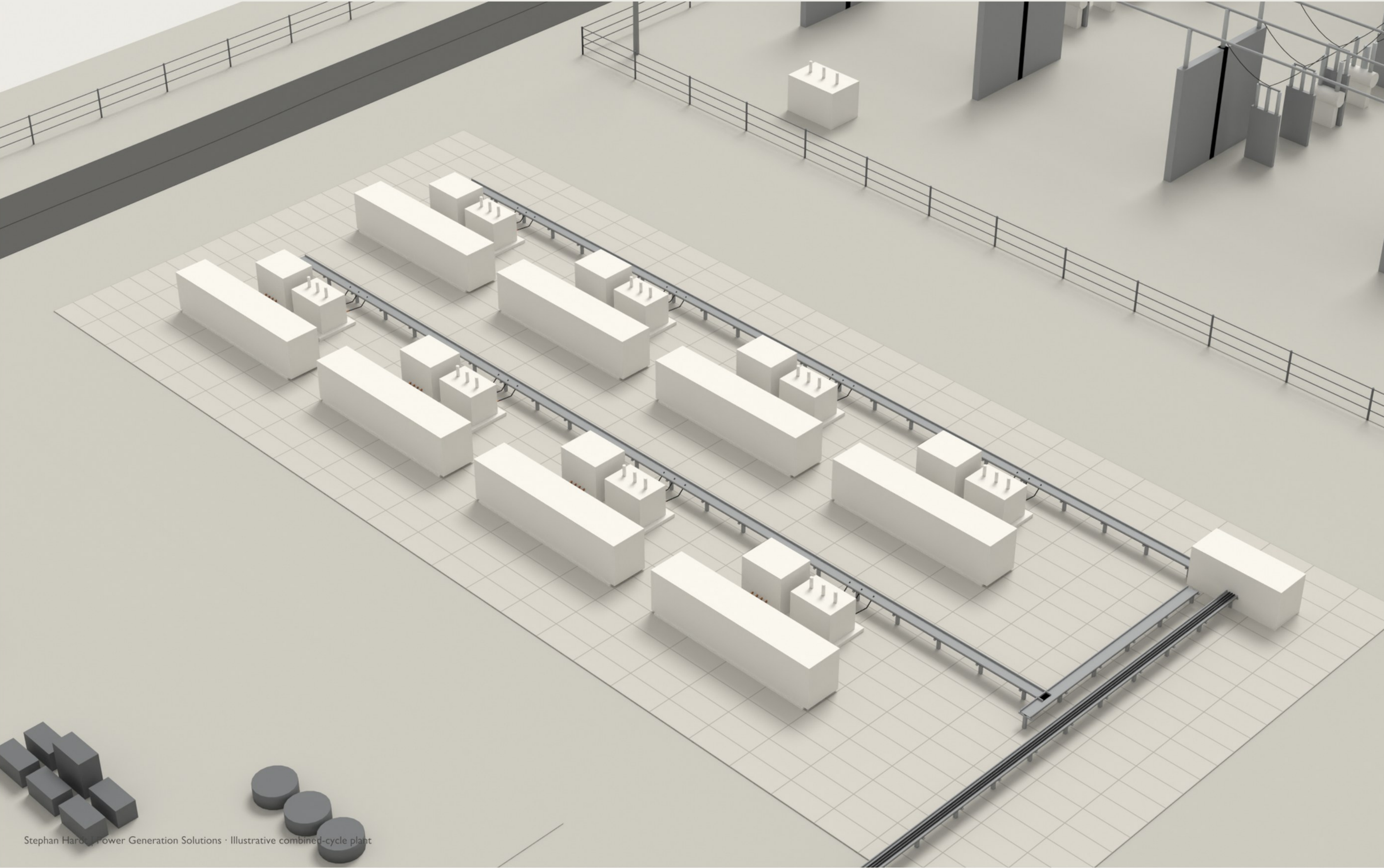


- A** Motor control centre lineup
- B** Variable-frequency drives
- C** UPS and battery rack
- D** DC distribution panel
- E** Overhead cable tray
- F** Cable terminations (copper)
- G** Cable entry pit from corridor lateral
- H** Fire-rated room partitions
- I** Roof HVAC

Explains where motor feeders, drive outputs and DC supplies are terminated, and how the entry pit feeds the overhead tray. Illustrative arrangement. Generic equipment and circuits; not procurement or construction data.

BESS yard

Energy storage | battery containers each paired with a PCS / transformer skid, collected to MV switchgear

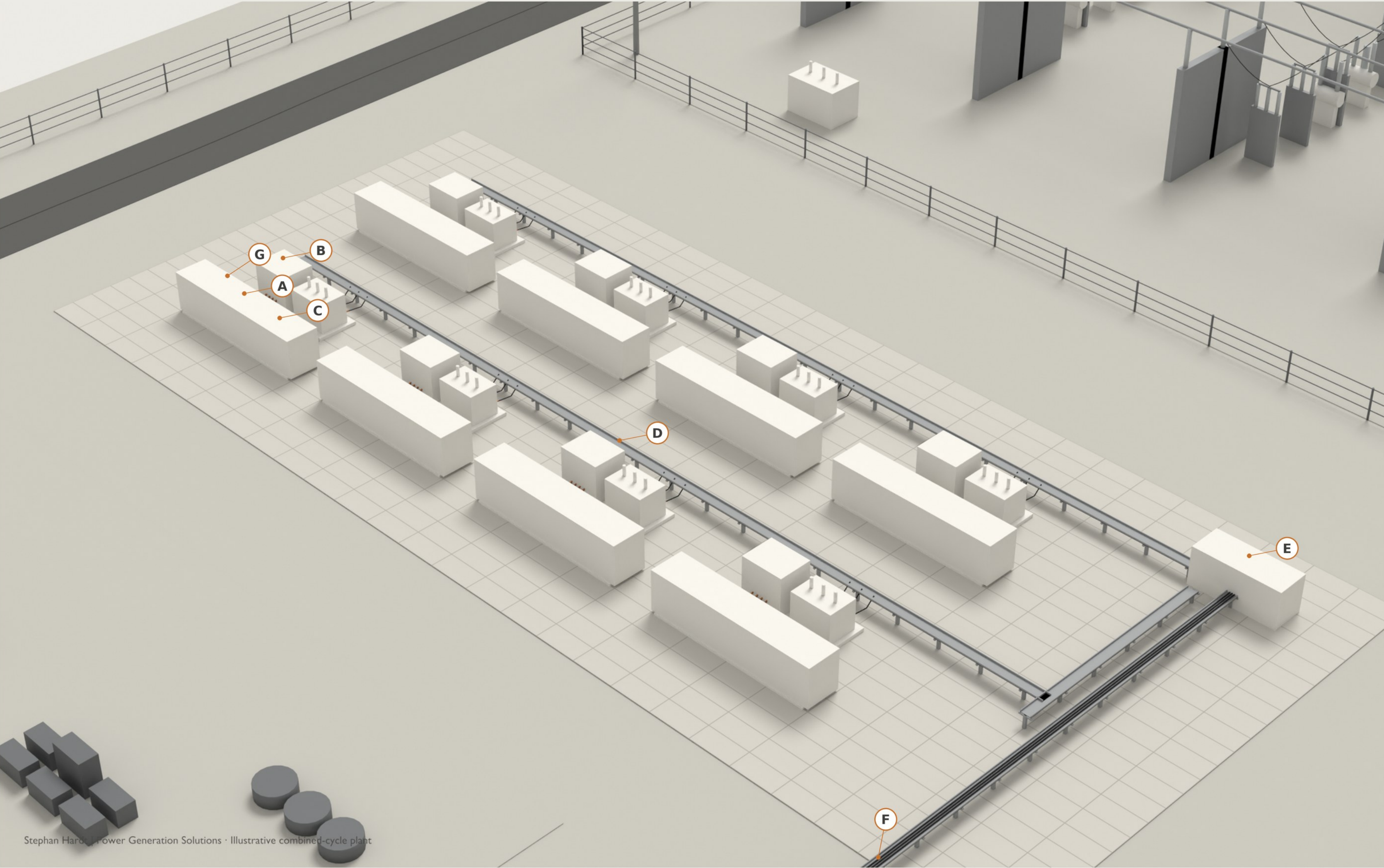


Stephan Harbeck Power Generation Solutions · Illustrative combined-cycle plant

Explains the DC-to-AC-to-MV chain: container to PCS, PCS to collector tray, collector to switchgear and one feeder to the e-house.
Illustrative arrangement. Generic equipment and circuits; not procurement or construction data.

BESS yard

Energy storage | battery containers each paired with a PCS / transformer skid, collected to MV switchgear

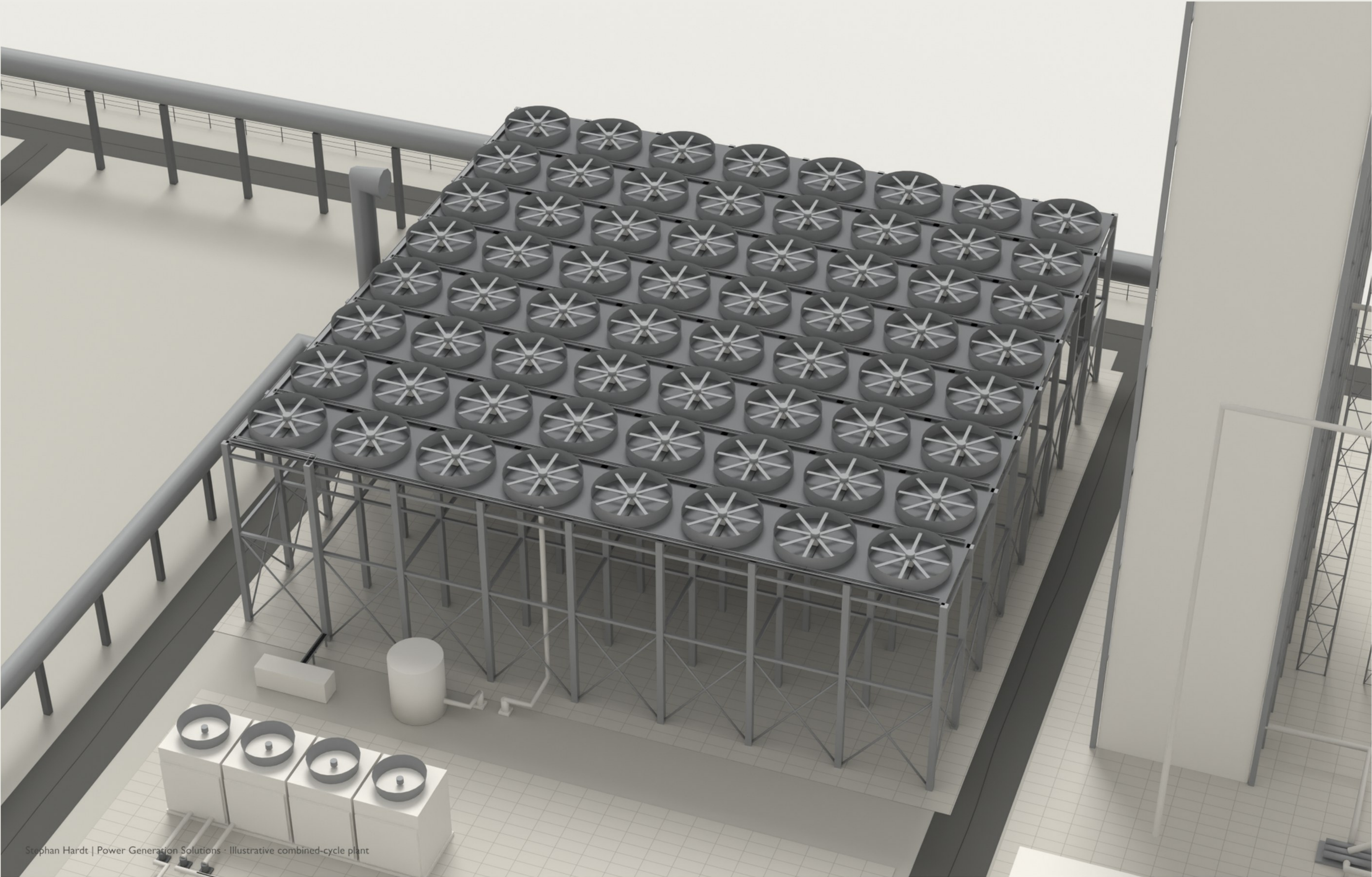


- A Battery enclosure (ISO container)
- B PCS inverter and MV transformer skid
- C DC cables, container to PCS
- D MV collector tray
- E BESS MV switchgear
- F MV feeder to e-house
- G Container HVAC

Explains the DC-to-AC-to-MV chain: container to PCS, PCS to collector tray, collector to switchgear and one feeder to the e-house. Illustrative arrangement. Generic equipment and circuits; not procurement or construction data.

Air-cooled condenser

Main condenser | fan cells, fan motors and their feeder cables, with the tube bundles and wind wall lifted off

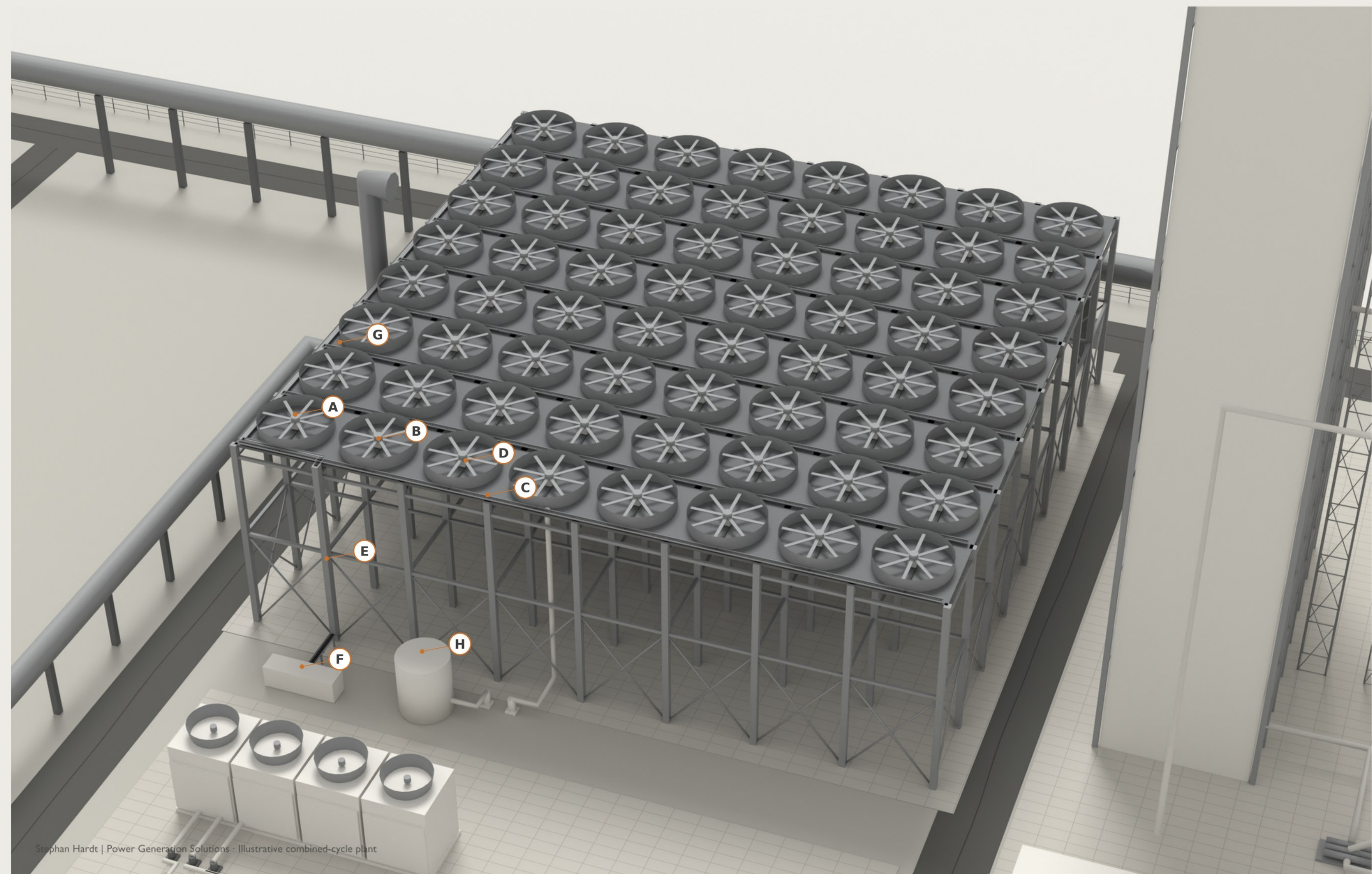


Stephan Hardt | Power Generation Solutions · Illustrative combined-cycle plant

Explains the electrical path to the sixty-four fan motors: MCC, riser, deck trays and one drop per motor, and where the steam duct arrives. Illustrative arrangement. Generic equipment and circuits; not procurement or construction data.

Air-cooled condenser

Main condenser | fan cells, fan motors and their feeder cables, with the tube bundles and wind wall lifted off



A ACC fan cell (8 x 8 array)

B Fan gearbox / motor

C Fan-deck feeder trays

D Fan motor feeders (copper terminations)

E Riser tray up the south face

F ACC motor control centre

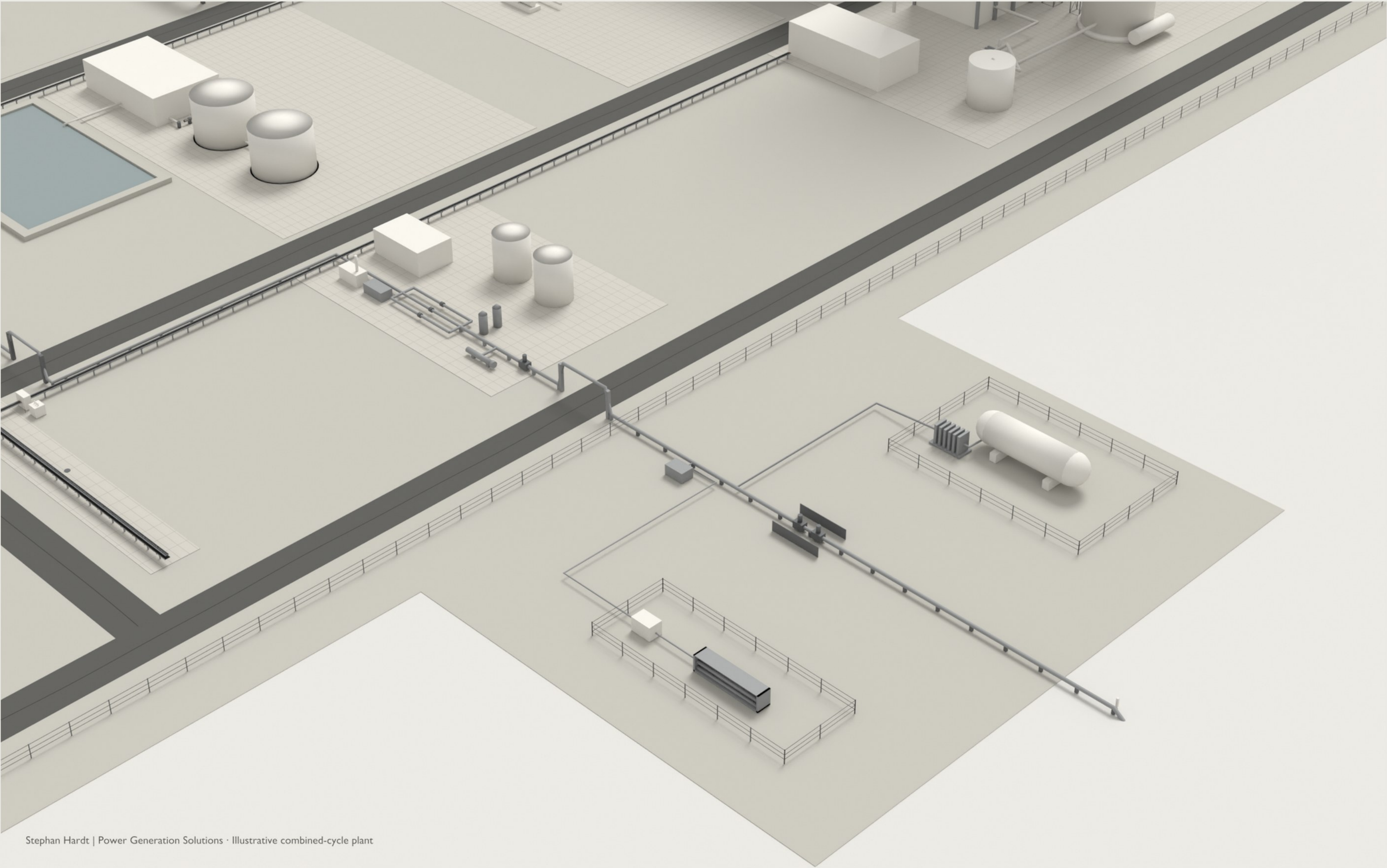
G Main steam duct from ST exhaust

H Condensate tank and pumps

Explains the electrical path to the sixty-four fan motors: MCC, riser, deck trays and one drop per motor, and where the steam duct arrives.
Illustrative arrangement. Generic equipment and circuits; not procurement or construction data.

Fuel gas and metering

Fuel supply | pipeline crossing, metering and conditioning, with LNG and hydrogen supply context outside the fence

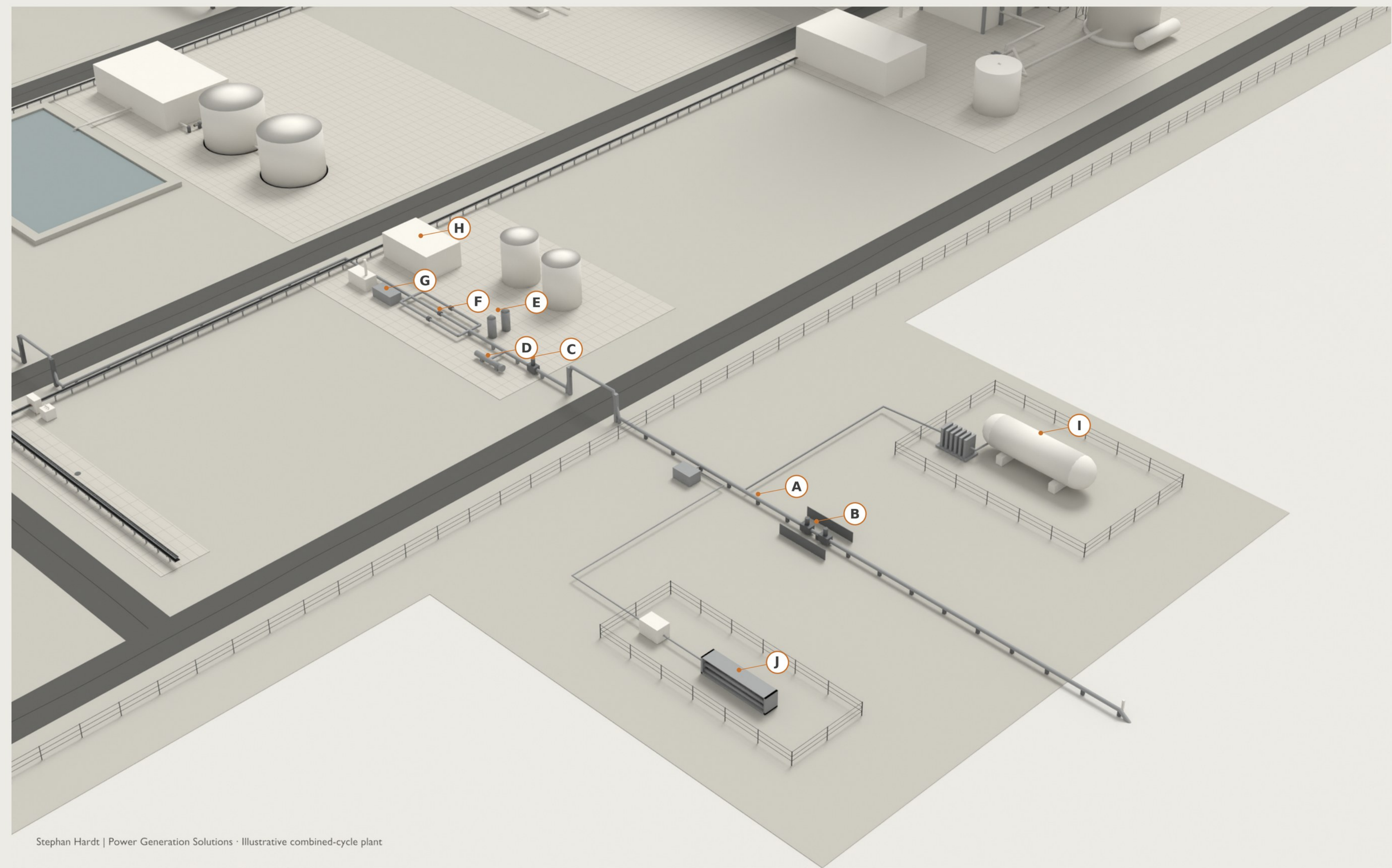


Stephan Hardt | Power Generation Solutions · Illustrative combined-cycle plant

Explains the primary pipeline supply through metering and pressure reduction, and where alternative LNG or hydrogen supply would tie in. Illustrative arrangement. Generic equipment and circuits; not procurement or construction data.

Fuel gas and metering

Fuel supply | pipeline crossing, metering and conditioning, with LNG and hydrogen supply context outside the fence

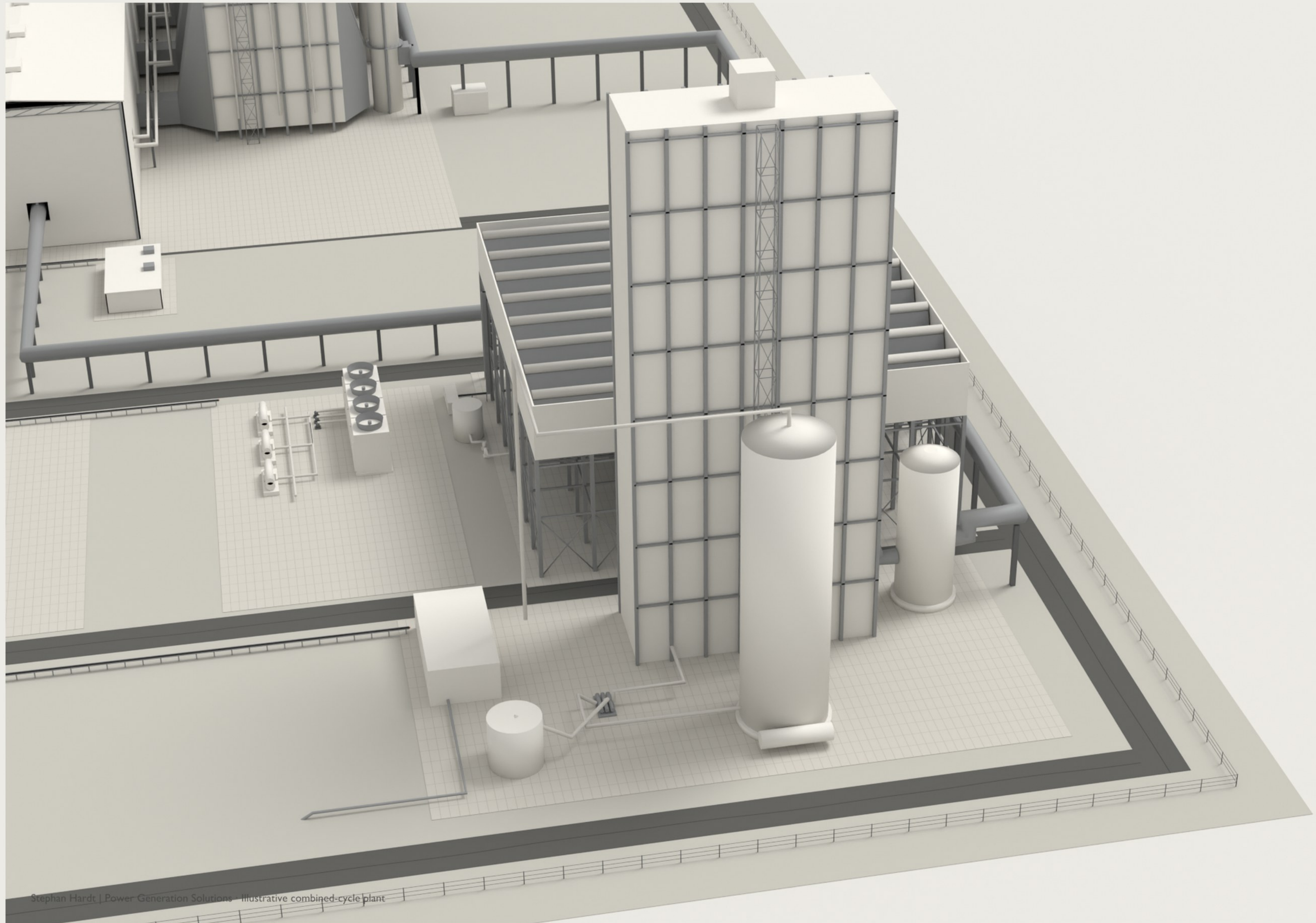


- Ⓐ Transmission pipeline supply (primary)
- Ⓑ Pipeline block-valve station
- Ⓒ Emergency shutdown valve
- Ⓓ Pig receiver
- Ⓔ Filter / separators
- Ⓕ Custody-transfer meter runs
- Ⓖ Pressure-reduction skid
- Ⓗ Gas metering / analyser house
- Ⓘ LNG storage vessel (alternative supply)
- ⓵ Hydrogen tube storage module (alternative / blend)

Explains the primary pipeline supply through metering and pressure reduction, and where alternative LNG or hydrogen supply would tie in. Illustrative arrangement. Generic equipment and circuits; not procurement or construction data.

Carbon capture block

Post-combustion capture | flue gas from the stacks through the DCC and absorber, solvent regeneration and CO2 compression

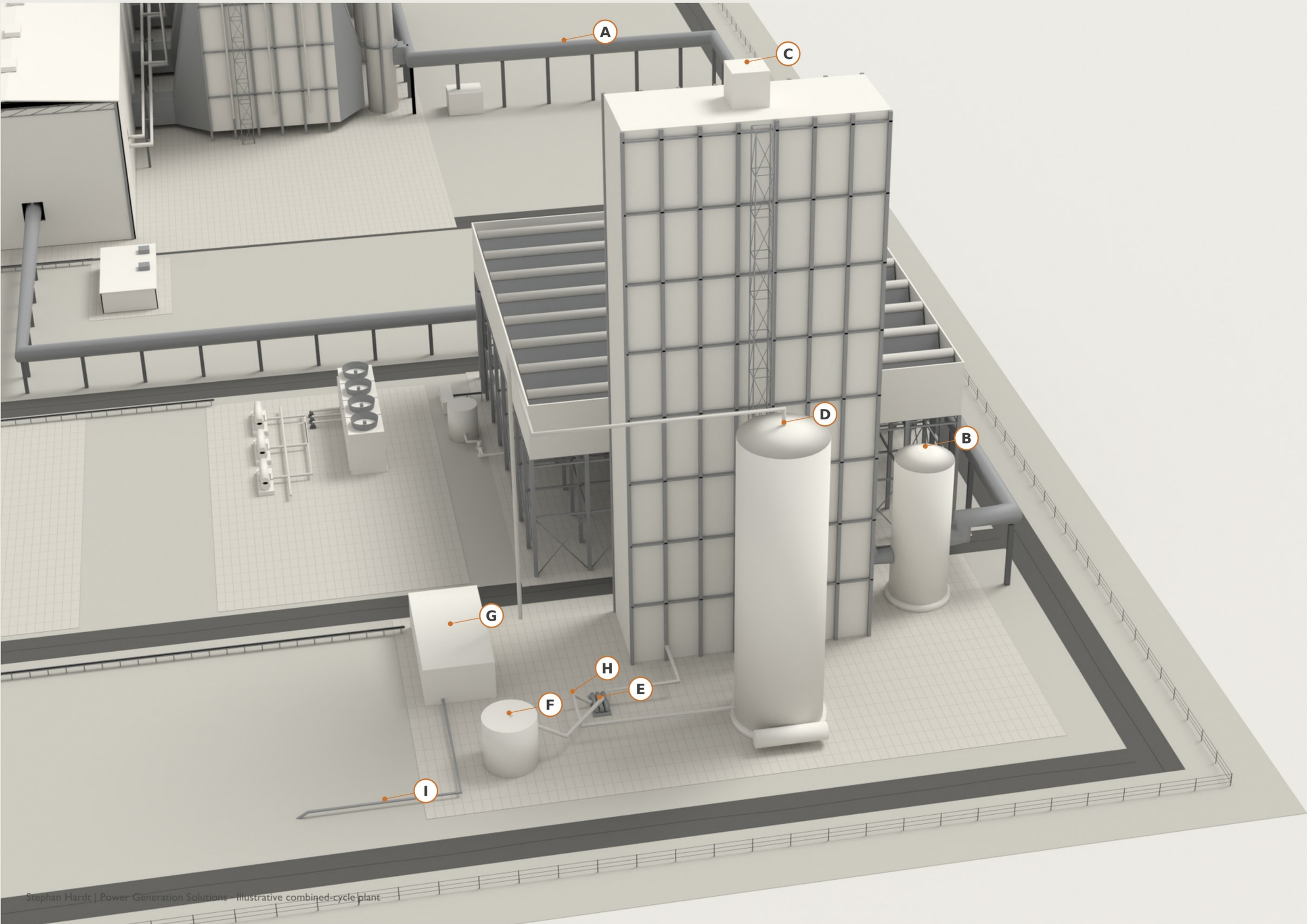


Stephan Hardt | Power Generation Solutions - Illustrative combined-cycle plant

Explains one representative capture train: quench, absorb, regenerate, compress and export; not a proven three-GT capture design. Illustrative arrangement. Generic equipment and circuits; not procurement or construction data.

Carbon capture block

Post-combustion capture | flue gas from the stacks through the DCC and absorber, solvent regeneration and CO2 compression



- A** Flue-gas collector duct from stacks

B Direct-contact cooler (flue-gas quench)

C CO2 absorber (amine) with treated-gas outlet
- D** Regenerator (stripper) and reboiler

E Lean / rich solvent heat exchanger

F Solvent storage tank
- G** CO2 compression and dehydration building

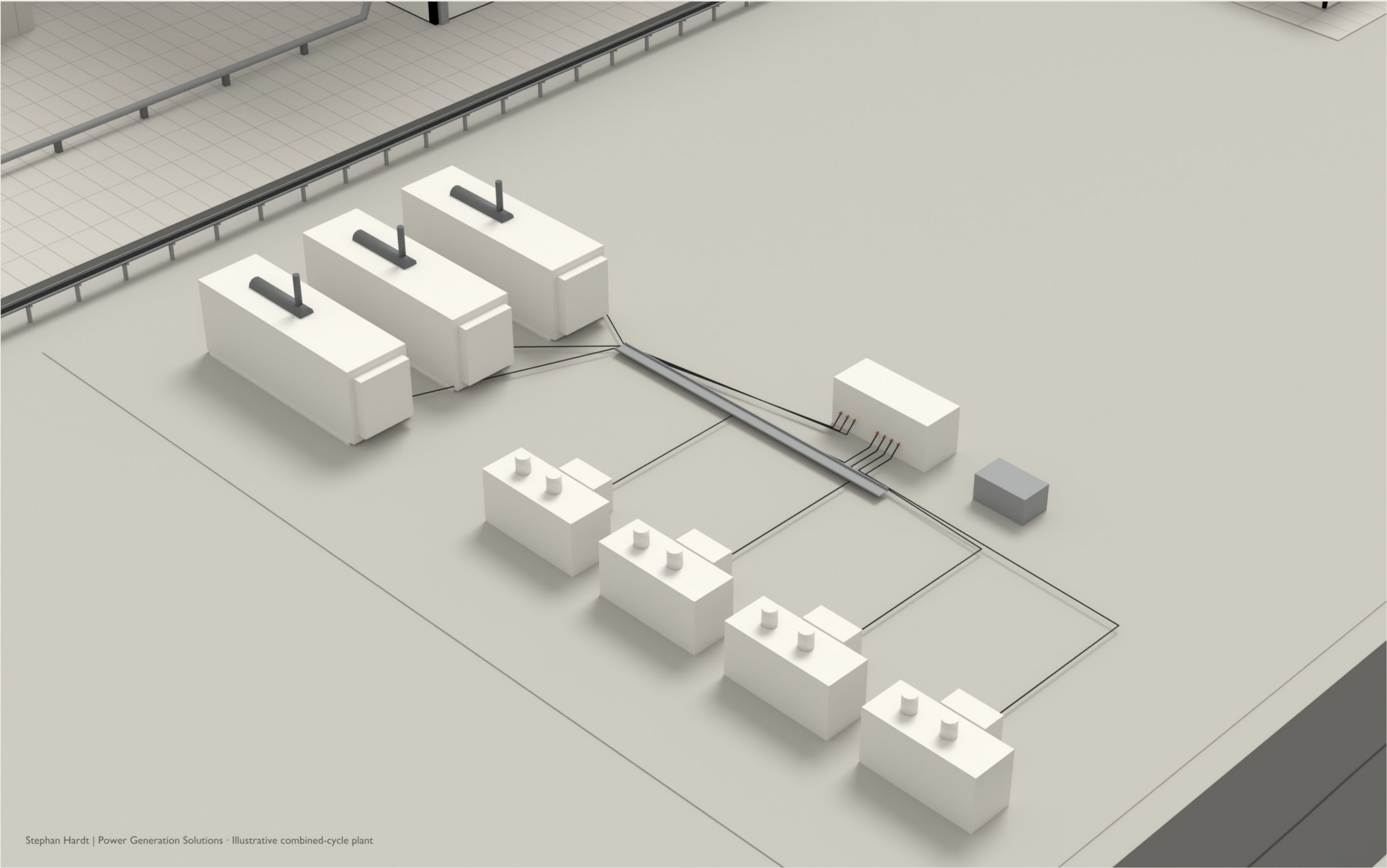
H Rich / lean solvent piping

I CO2 export pipeline (to storage, off view)

Explains one representative capture train: quench, absorb, regenerate, compress and export; not a proven three-GT capture design. Illustrative arrangement. Generic equipment and circuits; not procurement or construction data.

Modular power yard

Resilience | standby and black-start gensets, fuel-cell modules and the paralleling switchgear

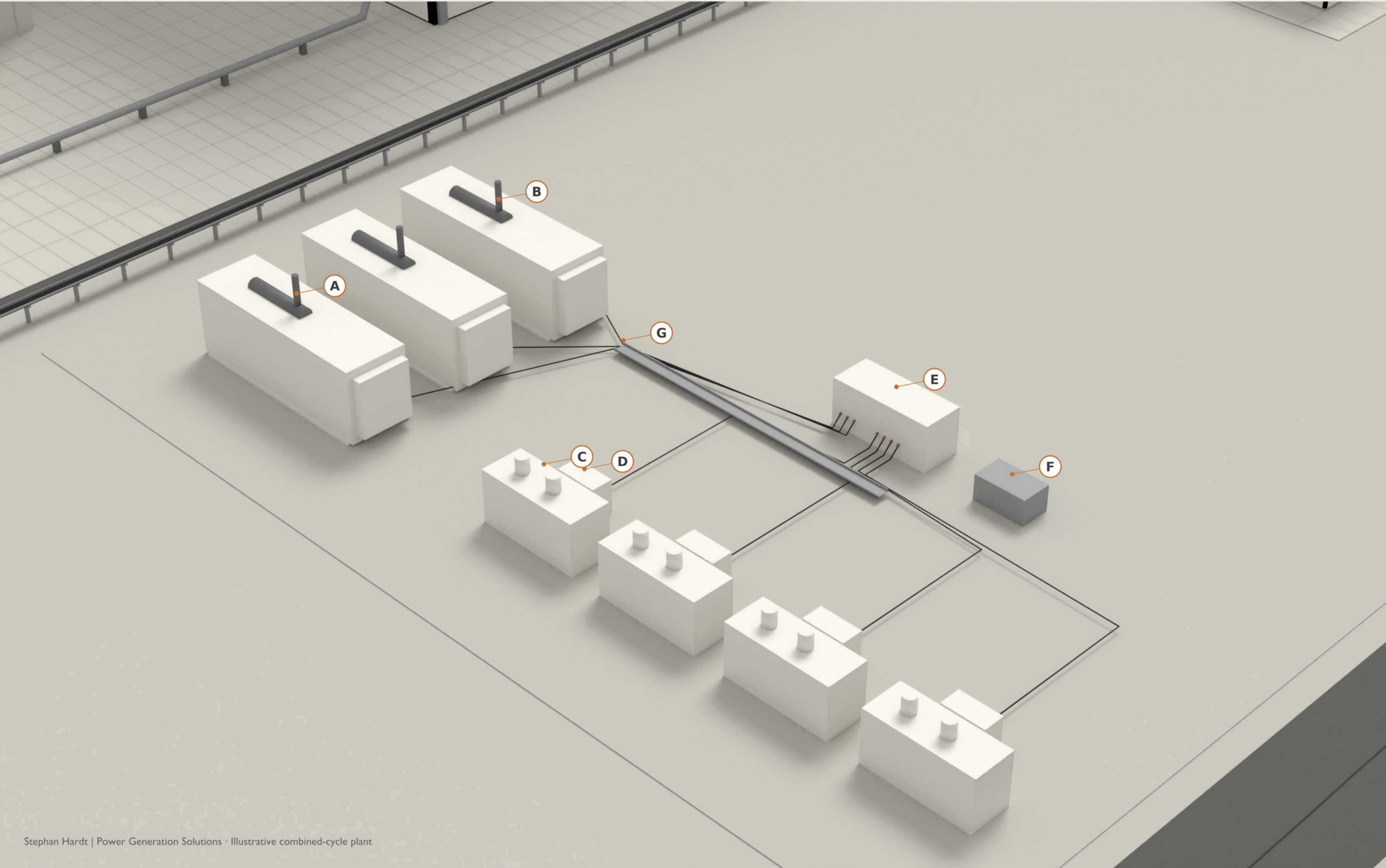


Stephan Hardt | Power Generation Solutions · Illustrative combined-cycle plant

Explains which sources can start the plant with the grid down and how they are paralleled onto the auxiliary system. Illustrative arrangement. Generic equipment and circuits; not procurement or construction data.

Modular power yard

Resilience | standby and black-start gensets, fuel-cell modules and the paralleling switchgear

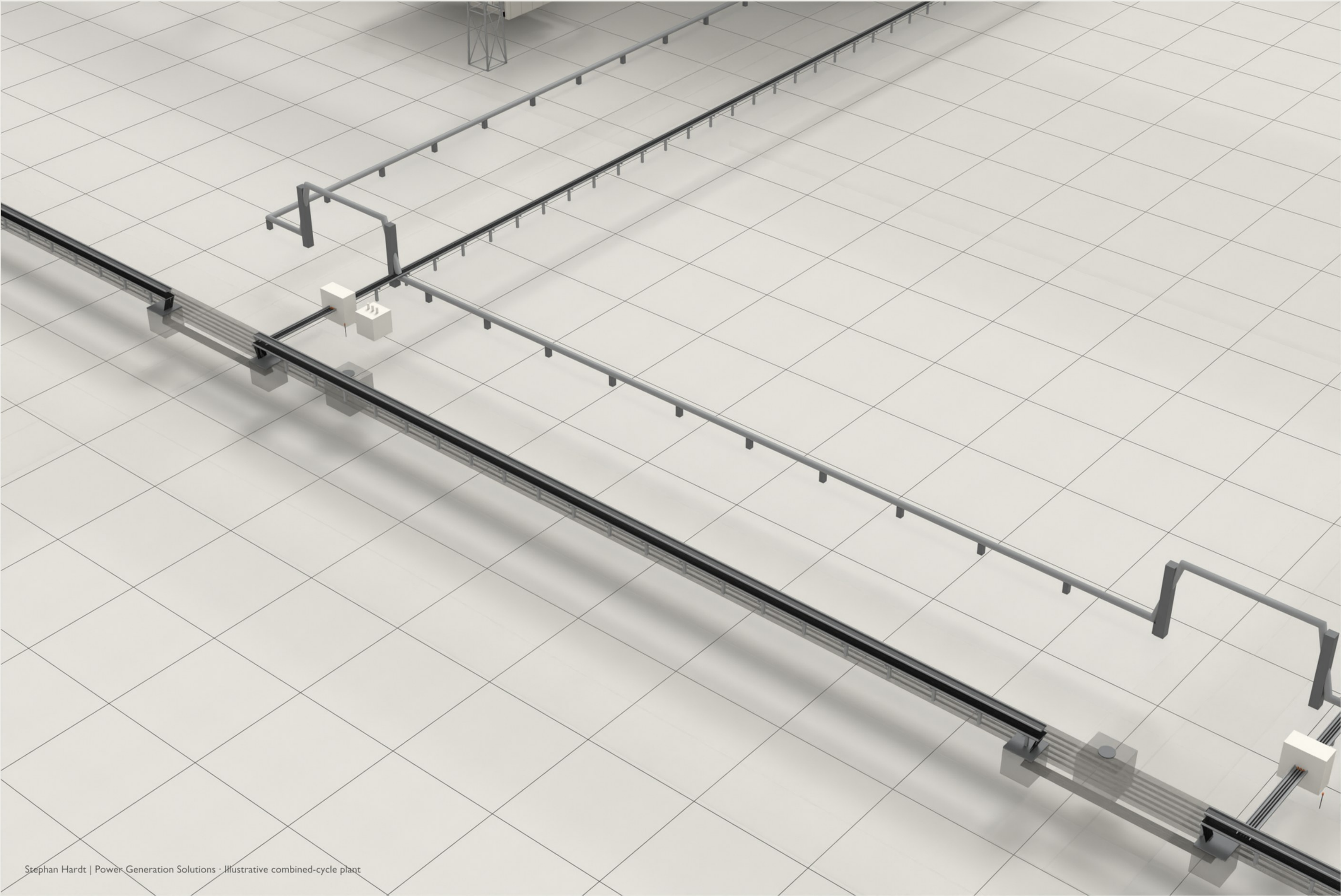


- A** Standby genset (containerised)
- B** Black-start genset (containerised)
- C** Fuel-cell modules with inverters
- D** Fuel-cell inverter skid
- E** Paralleling switchgear
- F** Fuel-gas conditioning skid
- G** Genset and fuel-cell output cables

Explains which sources can start the plant with the grid down and how they are paralleled onto the auxiliary system. Illustrative arrangement. Generic equipment and circuits; not procurement or construction data.

Site cable corridor

Cable routing | two-tier tray, duct bank below (X-ray), MV ring-main kiosks and a lateral to a building

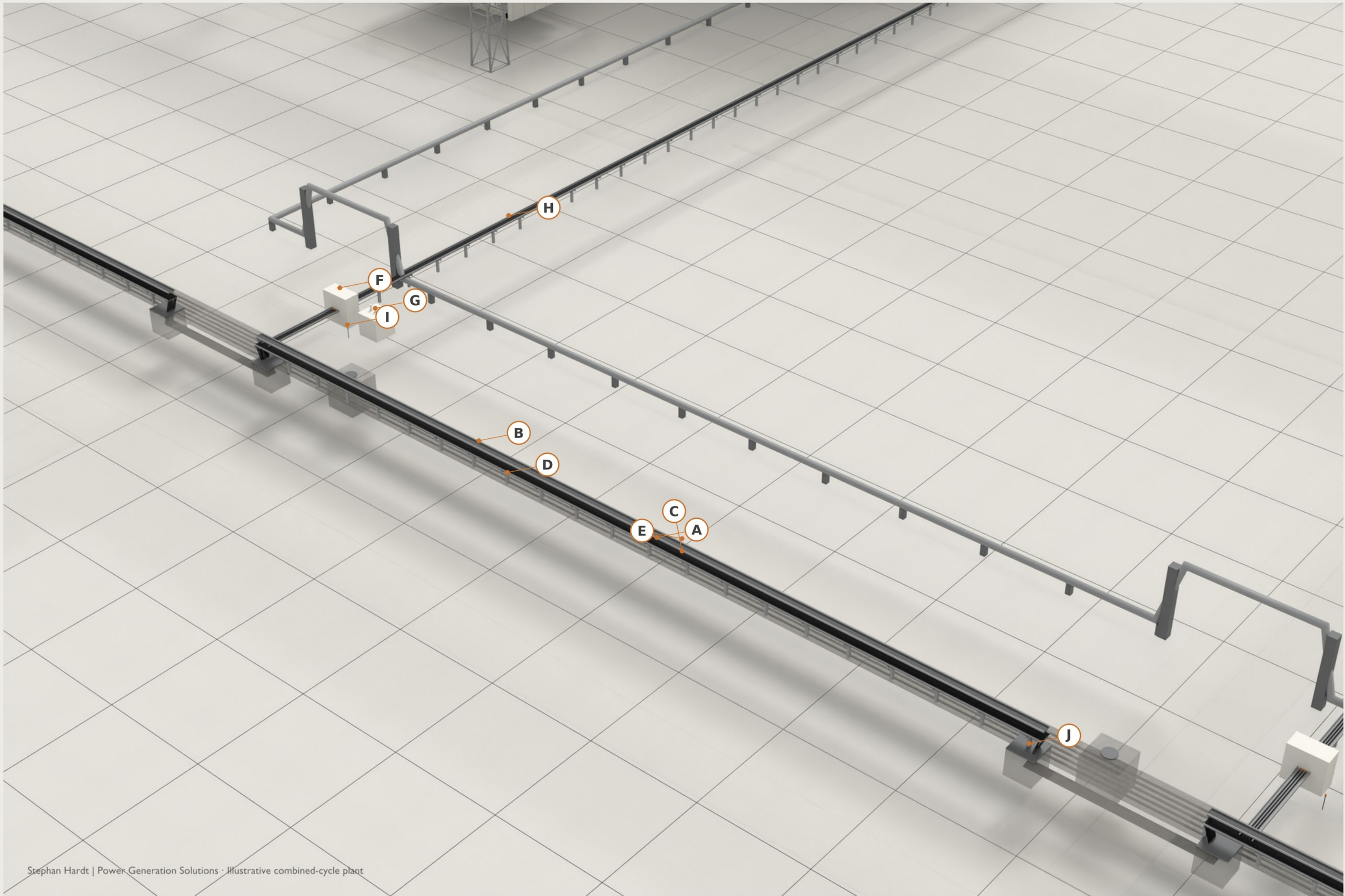


Stephan Hardt | Power Generation Solutions · Illustrative combined-cycle plant

Explains the separation of power (lower tier) from controls (upper tier), where cables go underground to cross a road, and how MV is tapped along the spine.
Illustrative arrangement. Generic equipment and circuits; not procurement or construction data.

Site cable corridor

Cable routing | two-tier tray, duct bank below (X-ray), MV ring-main kiosks and a lateral to a building

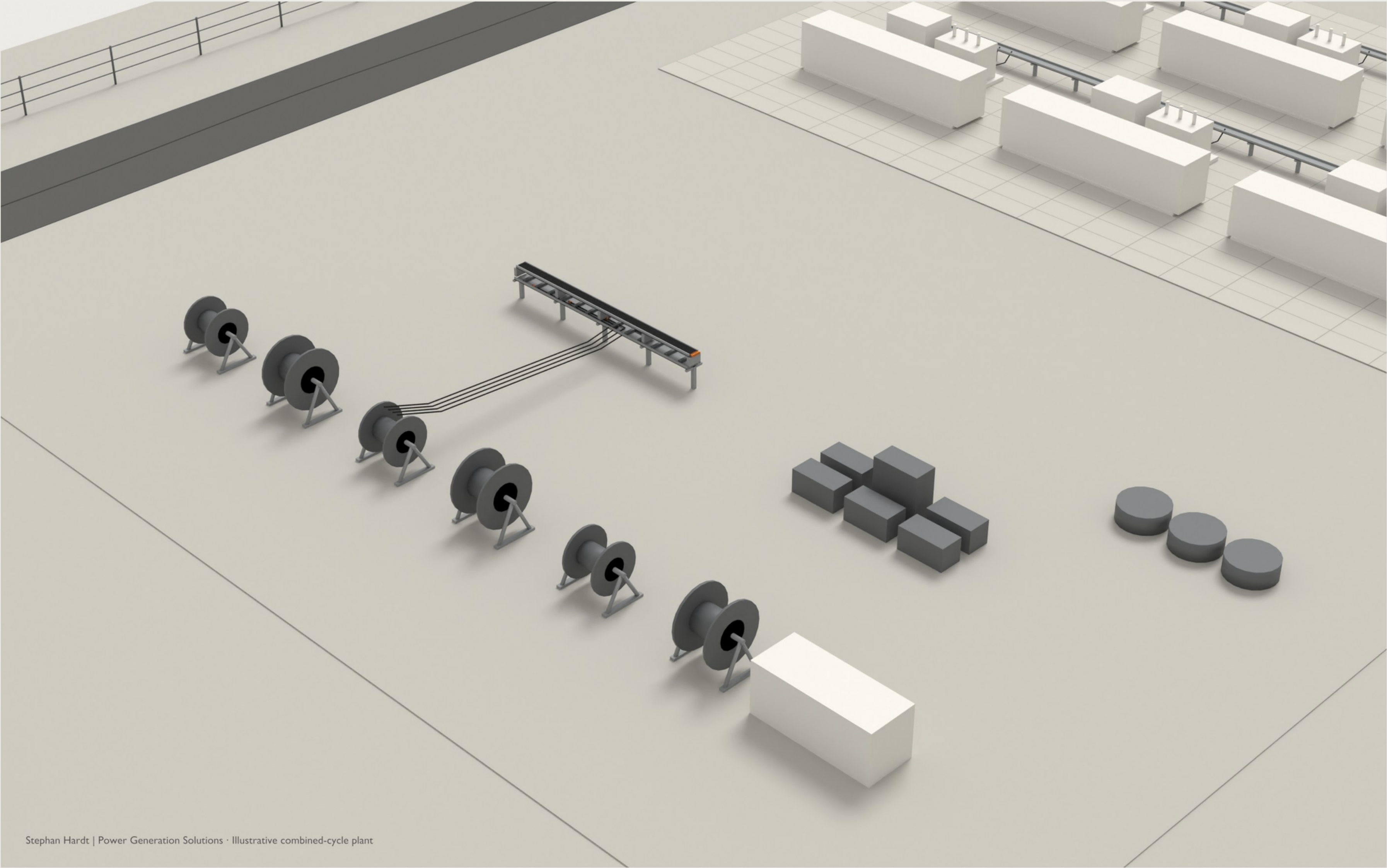


- A** Power tray (lower tier: MV / LV feeders)
- B** Control tray (upper tier: I&C, fibre)
- C** Concrete duct bank (below grade)
- D** Conduits with feeder cables (below grade)
- E** Pull-pit / manhole cover
- F** MV ring-main switchgear (MCC building)
- G** Pad-mount MV/LV transformer
- H** Lateral tray to building
- I** Grounding riser to earth grid (copper connector)
- J** Road crossing: cables dive into the duct bank

Explains the separation of power (lower tier) from controls (upper tier), where cables go underground to cross a road, and how MV is tapped along the spine.
Illustrative arrangement. Generic equipment and circuits; not procurement or construction data.

Reel staging and prefab spine

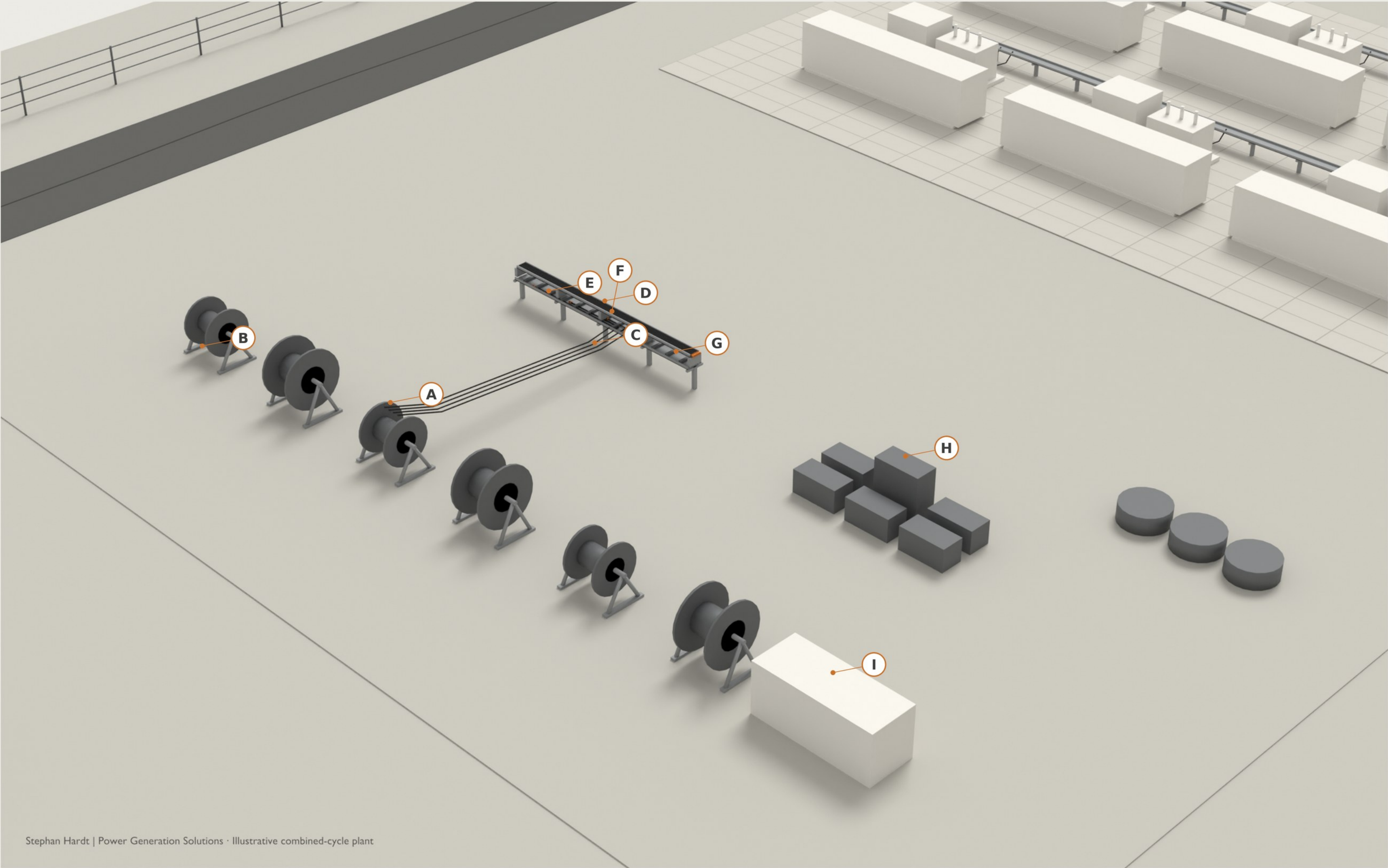
Installation | reels on payout stands feeding a prefabricated two-tier tray spine with cleats, bonding and end caps



Stephan Hardt | Power Generation Solutions · Illustrative combined-cycle plant

Reel staging and prefab spine

Installation | reels on payout stands feeding a prefabricated two-tier tray spine with cleats, bonding and end caps

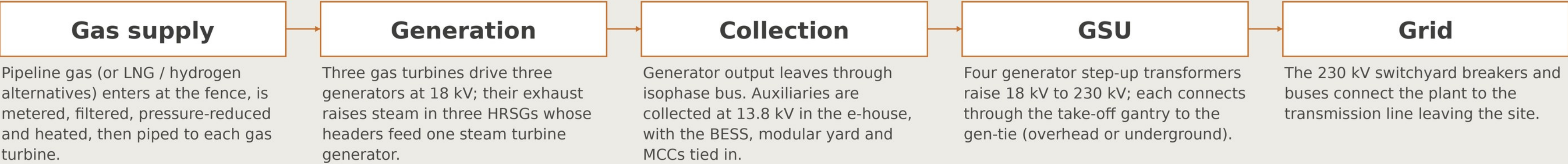


- | | | |
|--|---|----------------------------------|
| A Cable reels on payout stands | D Prefab two-tier tray spine (power below, controls above) | G Cable end caps (copper) |
| B Payout stand (A-frame, axle, brake) | E Cable cleats | H Crated equipment |
| C Cables paid out from reel | F Tray bonding jumpers (copper) | I Site office |

Explains how cable leaves the reel under control, is dressed into the spine and secured, and why the copper ends stay capped until termination. Illustrative arrangement. Generic equipment and circuits; not procurement or construction data.

How the plant connects

Power path | gas supply to generation to collection to GSU to grid



Inside the equipment

- Generator leads and terminal box
- Isophase bus and UAT tap
- Switchgear risers and terminations
- Fan motor drops on the ACC deck
- PCS to transformer skid links
- Cleats, bonding and end caps on the prefab spine

Across the site

- Two-tier tray corridor: power low, controls high
- Duct bank with manholes under each road crossing
- MV ring-main kiosks and pad-mount transformers
- Gen-tie towers or 230 kV duct bank to the switchyard
- Steam headers and main steam duct to the ACC
- Grounding grid with copper risers at each structure

Outside the view

- Transmission line beyond the switchyard
- Gas transmission pipeline and its block valves
- LNG deliveries and hydrogen supply
- CO2 export pipeline to storage
- Make-up water source and wastewater discharge
- Control system networks between buildings

Abbreviations

ACC	Air-cooled condenser	GSU	Generator step-up transformer	IPB	Isophase bus	UAT	Unit auxiliary transformer
BESS	Battery energy storage system	GT	Gas turbine	MCC	Motor control centre	UPS	Uninterruptible power supply
CCS	Carbon capture and storage	HRSG	Heat recovery steam generator	PCS	Power conversion system	VFD	Variable-frequency drive
DCC	Direct-contact cooler	HV / MV / LV	High / medium / low voltage	ST	Steam turbine	XLPE	Cross-linked polyethylene (cable insulation)